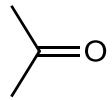
| | | |
|---|---|---|
| 1(a)(i) | (halogens can) remove electron(s) (from a reactant) | **1** |

| | | |
|---|---|---|
| 1(a)(ii) | $2Al + 3Cl_2 \rightarrow 2AlCl_3$ | **1** |

# Examiner report missing
# 9701_w25_qp_24 Question 1

| | | |
|---|---|---|
| 2(a) | **M1** $Al_2O_3$ is ionic<br><br>**M2** $P_4O_{10}$ is a (simple) molecule<br><br>**M3** ionic bonds are stronger than intermolecular forces | **3** |
| 2(c)(i) | increases rate by providing a different mechanism with a lower activation energy | **1** |
| 2(c)(ii) | $2CH_3OH \rightarrow CH_3OCH_3 + H_2O$ | **1** |
| 2(c)(iii) | $P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$ | **1** |

# Examiner report missing
# 9701_w25_qp_24 Question 2

| | | |
|---|---|---|
| 3(a)(i) |  | **1** |
| 3(b)(i) | $(CH_3)_2CHOH + Na \rightarrow (CH_3)_2CHO^{(-)}Na^{(+)} + \frac{1}{2}H_2$ | **1** |
| 3(b)(iv) | $C_6H_{14}O + 9O_2 \rightarrow 6CO_2 + 7H_2O$ | **1** |

# Examiner report missing
# 9701_w25_qp_24 Question 3

| | | |
|---|---|---|
| 4(a)(i) | **M1** (molecules with the) same molecular formula / same number of atoms of each element<br><br>**M2** different structural formulae/structures | **2** |
| 4(a)(iv) | **M1** $CH_3COCH_3$ + $CH_3COOH$ as products<br><br>**M2** $(CH_3)_2C = CHCH_3 + 3[O] \rightarrow CH_3COCH_3 + CH_3COOH$ | **2** |
| 4(b)(i) | KCN in ethanol (and heat) | **1** |
| 4(b)(ii) | $(C_2H_5)_2CHCN + 2H_2O + HCl \rightarrow (C_2H_5)_2CHCOOH + NH_4Cl$ | **1** |

# Examiner report missing
# 9701_w25_qp_24 Question 4

| | | |
|---|---|---|
| 1(a)(i) | 24 protons 28 neutrons 24 electrons | **1** |
| 1(a)(iii) | (relative) abundance (of each isotope) | **1** |
| 1(a)(iv) | The fourth isotope has any value of RIM less than 51.996 | **1** |
| 1(c)(i) | Cr / $Cr_2O_7^{2-}$ **AND** it gains electrons | **1** |
| 1(c)(iii) | *Due to an issue with question 1(c)(iii), the question has been removed from the question paper.* | **N/A** |
| 1(d)(iii) | **M1** (mixture becomes more) orange<br>**M2** forward reaction is endothermic<br>**M3** (mixture becomes more) orange<br>**M4** equilibrium moves (forwards/right) to compensate for extra acid / $H^+$ | **4** |
| 1(e) | **M1** hydrolysis<br>**M2** $CrF_4 + 2H_2O \rightarrow CrO_2 + 4HF$ | **2** |

# Examiner report missing
# 9701_w25_qp_23 Question 1

| | | |
|---|---|---|
| 2(a)(i) | **M1** nuclear charge increases (across period) **AND** similar shielding<br><br>**M2** greater / stronger attraction to nucleus | **2** |

| | | |
|---|---|---|
| 2(c)(i) | $Na_2O + 2HCl \rightarrow 2NaCl + H_2O$ | **1** |

| | | |
|---|---|---|
| 2(c)(ii) | $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$ | **1** |

# Examiner report missing
# 9701_w25_qp_23 Question 2

| | | |
|---|---|---|
| 3(a) | simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule | **1** |

# Examiner report missing
# 9701_w25_qp_23 Question 3

| | | |
|---|---|---|
| 1(a)(i) | 25 protons 30 neutrons 25 electrons | **1** |
| 1(a)(ii) | (atoms of an element) with the same number of protons and different number of neutrons | **1** |
| 1(a)(iii) | **M1** (relative) isotopic mass.<br>**OR** (relative isotopic) mass of Mn-55 and Mn-53<br><br>**M2** (relative) abundance of each isotope | **2** |
| 1(c)(i) | Mn / $MnO_2$ **AND** it gains electrons | **1** |
| 1(c)(iii) | *Due to an issue with question 1(c)(iii), the question has been removed from the question paper.* | **N/A** |

# Examiner report missing
# 9701_w25_qp_22 Question 1

| | | |
|---|---|---|
| 2(a)(i) | delocalised electrons move (through the structure / between layers) | **1** |
| 2(d)(i) | $2Al + 3Cl_2 \rightarrow 2AlCl_3$ | **1** |
| 2(d)(ii) | $PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$ | **1** |
| 2(e)(iii) | **M1** (mixture / solution becomes more) yellow<br><br>**M2** (forward) reaction is exothermic (equilibrium shifts left)<br>**OR** equilibrium / reaction shifts (left) to absorb (additional) heat / energy<br><br>**M3** (mixture / solution becomes more) red<br><br>**M4** fewer moles of gas on right-hand side<br>**OR** (gas on left and) no gas on the right | **4** |
| 2(f)(i) | $MgCl_2(aq) + Na_2CO_3(aq) \rightarrow MgCO_3(s) + 2NaCl(aq)$<br>**M1** formulae in balanced equation<br>**M2** state symbols | **2** |

# Examiner report missing
# 9701_w25_qp_22 Question 2

| | | |
|---|---|---|
| 3(a) | **M1** (family of molecules / compounds / series with) same general formula<br>**OR** (family of molecules where) molecular formula differs (only) by (units of) $CH_{2(n)}$.<br><br>**M2** similar chemical properties / reactions | **2** |

| | | |
|---|---|---|
| 3(e)(iii) | $C_6H_{14} + 9½O_2 \rightarrow 6CO_2 + 7H_2O$ | **1** |

# Examiner report missing
# 9701_w25_qp_22 Question 3

| | | |
|---|---|---|
| 4(a)(i) | free-radical substitution | **1** |
| 4(c)(i) | orange to green | **1** |
| 4(c)(ii) | $CH_3CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2COOH + H_2O$ | **1** |

# Examiner report missing
# 9701_w25_qp_22 Question 4

| | | |
|---|---|---|
| 1(a)(i) | 24 protons 28 neutrons 24 electrons | **1** |
| 1(a)(iii) | (relative) abundance (of each isotope) | **1** |
| 1(a)(iv) | The fourth isotope has any value of RIM less than 51.996 | **1** |
| 1(c)(i) | Cr / $Cr_2O_7^{2-}$ **AND** it gains electrons | **1** |
| 1(c)(iii) | *Due to an issue with question 1(c)(iii), the question has been removed from the question paper.* | **N/A** |
| 1(d)(iii) | **M1** (mixture becomes more) orange<br>**M2** forward reaction is endothermic<br>**M3** (mixture becomes more) orange<br>**M4** equilibrium moves (forwards/right) to compensate for extra acid / $H^+$ | **4** |
| 1(e) | **M1** hydrolysis<br>**M2** $CrF_4 + 2H_2O \rightarrow CrO_2 + 4HF$ | **2** |

# Examiner report missing
# 9701_w25_qp_21 Question 1

| | | |
|---|---|---|
| 2(a)(i) | **M1** nuclear charge increases (across period) **AND** similar shielding<br><br>**M2** greater / stronger attraction to nucleus | **2** |

| | | |
|---|---|---|
| 2(c)(i) | $Na_2O + 2HCl \rightarrow 2NaCl + H_2O$ | **1** |

| | | |
|---|---|---|
| 2(c)(ii) | $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$ | **1** |

# Examiner report missing
# 9701_w25_qp_21 Question 2

| | | |
|---|---|---|
| 3(a) | simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule | **1** |

# Examiner report missing
# 9701_w25_qp_21 Question 3

| | | |
|---|---|---|
| 1(a) | <table><tr><th>*Mg*</th><th>*Si*</th><th>*P*</th></tr><tr><td>**M1** Giant metallic</td><td>**M2** Giant molecular</td><td>**M3** Simple molecular</td></tr></table> | **3** |
| 1(c)(i) | $Ar^{2+}(g) \rightarrow Ar^{3+}(g) + e^{(-)}$<br><br>**M1** a correct equation which represents an ionisation process *e.g.* $X^{n+} \rightarrow X^{(n+1)+} + e^{(-)}$<br>**M2** correct equation to show third ionisation energy of Ar including (g) state symbols | **2** |
| 1(d)(i) | $P_4 + 5O_2 \rightarrow P_4O_{10}$ | **1** |
| 1(d)(ii) | $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$ | **1** |

# Examiner report missing
# 9701_s25_qp_24 Question 1

| | | |
|---|---|---|
| 3(a)(i) | 0.00658(0) mol | **1** |

| | | |
|---|---|---|
| 3(a)(ii) | **M1** $I_2$ in 25.0 $cm^3$ 0.00658 / 2 = 0.00329 (mol)<br>**M2** 0.00329 × 1000 / 25.0 = 0.132 (mol)<br><br>alternative answer using $9.42 \times 10^{-3}$:<br>**M1** 0.00471<br>**M2** 0.1884 | **2** |

| | | |
|---|---|---|
| 3(b) | **M1** 0.2 – 0.154 = 0.046 (mol $H_2$ and / or $I_2$ reacts)<br>**M2** mol $H_2$ at equilibrium 0.1 – 0.046 = 0.054<br>**M3** mol HI at equilibrium 0.3 + (0.046 × 2) = 0.392 | **3** |

# Examiner report missing
# 9701_s25_qp_24 Question 3

| | | |
|---|---|---|
| 4(a) | $C_5H_{10}O_2$ | **1** |
| 4(b) | propyl ethanoate | **1** |
| 4(c)(i) | condensation | **1** |
| 4(d)(i) | $CH_3CN + HCl + 2H_2O \rightarrow CH_3COOH + NH_4Cl$ | **1** |

# Examiner report missing
# 9701_s25_qp_24 Question 4

| | | |
|---|---|---|
| 6(a)(i) | simplest (whole number) ratio of atoms (of the elements) in a compound | **1** |

| | | | | | |
|---|---|---|---|---|---|
| 6(a)(ii) | | C | : | H | **2** |
| | **M1** | $\frac{85.7}{12}$ or 7.14 | : | $\frac{14.3}{1}$ or 14.3 | |
| | | 1 | : | 2 | |
| | **M2** | $CH_2$ | | | |

# Examiner report missing
# 9701_s25_qp_24 Question 6

| | | |
|---|---|---|
| 2(a) | $Na_2O$ | **1** |
| 2(b) | $P_4O_{10} + 12NaOH \rightarrow 4Na_3PO_4 + 6H_2O$ | **1** |
| 2(e)(i) | $NaAl(OH)_4$ | **1** |
| 2(e)(ii) | $Al_2(SO_4)_3$ | **1** |

# Examiner report missing
# 9701_s25_qp_23 Question 2

| | | |
|---|---|---|
| 4(a)(i) | Check final graph for range of acceptable answers<br>● $\pm(1–0.5) / 100 = \pm 0.005$<br>● 0<br>● mol $dm^{-3}$ $s^{-1}$ for both<br><br>Any 2 points = 1 mark, all 3 points = 2 marks | **2** |

| | | |
|---|---|---|
| 4(a)(ii) | $Na_2S_2O_3$ because reaction stops before all the HC$l$ is used up.<br>**OR** $Na_2S_2O_3$ because there is still HC$l$ remaining<br>(when the reaction has stopped) | **1** |

| | | |
|---|---|---|
| 4(a)(iv) | M1 0.001(00) mol sulfur produced<br>M2 M1 $\times$ 6.02 $\times$ $10^{23}$<br>= 6.02 $\times$ $10^{20}$ | **2** |

# Examiner report missing
# 9701_s25_qp_23 Question 4

| | | |
|---|---|---|
| 6(a)(i) | two or more reactants / molecules react to make (only) one product | **1** |
| 6(c)(i) | KCN/potassium cyanide in ethanol **AND** heat | **1** |
| 6(c)(ii) | (acid) hydrolysis | **1** |
| 6(c)(iii) | $HOCH_2CH_2CN + 2H_2O + HCl \rightarrow HOCH_2CH_2COOH + NH_4Cl$ | **1** |

# Examiner report missing
# 9701_s25_qp_23 Question 6

| | | |
|---|---|---|
| 1(a)(i) | (both) giant molecular | **1** |

| | | |
|---|---|---|
| 1(a)(iii) | no delocalised electrons (to carry charge through the structure)<br>**OR** <u>all</u> (4) outer electrons are used in bonding | **1** |

| | | |
|---|---|---|
| 1(b)(i) | $PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$ | **1** |
| | $SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$ | **1** |

# Examiner report missing
# 9701_s25_qp_22 Question 1

| | | |
|---|---|---|
| 2(a)(i) | **route 1**<br>**M1** average (weighted) mass of (an) atom(s) (of an element)<br>**OR** average (weighted) mass of the isotope(s) (of an element)<br><br>**M2** compared to the mass of the unified atomic mass unit<br>**OR** on a scale in which a carbon-12 atom / isotope has a mass of exactly 12 (units)<br>**OR** on a scale in which the mass of a carbon-12 atom/isotope is exactly 12 (units)<br>**OR** compared to $\frac{1}{12}$ mass of one atom of C-12<br><br>**route 2**<br>**M1** average mass of one mol of isotopes / atoms (of an element)<br><br>**M2** relative / compared to $\frac{1}{12}$ of the mass of one mol of C-12 atoms/isotopes<br>**OR** when the mass of one mol C-12 atoms / isotopes is exactly 12 (000) g | **2** |

| | | |
|---|---|---|
| 2(a)(ii) | **M1** relative abundance of unknown isotope = (100 – 91.9 – 6.0) = 2.1%<br><br>**M2** 55.8 = (53.9 × 6.0 + 55.9 × 91.9 + ? × M1) / 100 = 56.9 | **2** |

# Examiner report missing
# 9701_s25_qp_22 Question 2

| | | |
|---|---|---|
| 3(a)(i) | two (or more) reactants / molecules react to make (only) one product | **1** |

| | | |
|---|---|---|
| 3(b)(i) | $H_2$ / hydrogen **AND** some X / alkene left(over) / remains (at the end of this reaction) | **1** |

# Examiner report missing
# 9701_s25_qp_22 Question 3

| | | |
|---|---|---|
| 4(a)(i) | $LiAlH_4$ | **1** |

| | | |
|---|---|---|
| 4(b)(i) | substitution | **1** |

| | | |
|---|---|---|
| 4(b)(iii) | $CH_3CH_2CH_2CN + NaOH + H_2O \rightarrow CH_3CH_2CH_2CO_2Na + NH_3$<br>**OR**<br>$CH_3CH_2CH_2CN + NaOH + 2H_2O \rightarrow CH_3CH_2CH_2CO_2Na + NH_4OH$<br><br>**M1** correct organic product (as molecule or anion)<br>**M2** rest of equation is correct | **2** |

# Examiner report missing
# 9701_s25_qp_22 Question 4

| | | |
|---|---|---|
| 5(a) | $C_3H_4O_3$ | **1** |

| | | |
|---|---|---|
| 5(b)(i) | $5.00 \times 10^{-3} \times 36.65 / 1000$<br>$= 1.833 \times 10^{-4} / 1.8325 \times 10^{-4}$ | **1** |

| | | |
|---|---|---|
| 5(b)(ii) | **M1** expression and / or correct calculation to show how the amount of vitamin C in 150 g of lemon peel is calculated<br>$1.8325 \times 10^{-4}$ mol $\times 176 \times 4$<br><br>**M2** correct calculation based on (**M1** / 150) $\times$ 100 = 0.086(%)<br>Using $1.8325 \times 10^{-4}$ for **M1**, **M2** = 0.086 **OR** 0.087(%)<br><br>Alternative answer using $7.65 \times 10^{-4}$:<br>**M1** $7.65 \times 10^{-4} \times 176 \times 4 = 0.5386 / 0.539$<br>**M2** correct calculation based on (**M1** / 150) $\times$ 100 = 0.359 / 0.36 (%) | **2** |

# Examiner report missing
# 9701_s25_qp_22 Question 5

| | | |
|---|---|---|
| 1(a)(i) | sodium – metallic<br>**AND** sodium oxide – ionic | **1** |
| 1(a)(iii) | (sodium and oxide) ions move (through the liquid) | **1** |
| 1(b)(i) | $2Na + 2H_2O \rightarrow 2NaOH + H_2$ | **1** |

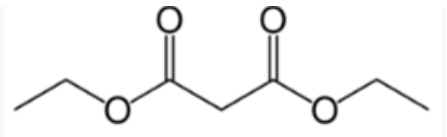
# Examiner report missing
# 9701_s25_qp_21 Question 1

| | | |
|---|---|---|
| 4(a) | $C_3H_4O_3$ | **1** |

| | | |
|---|---|---|
| 4(b)(i) | $(28.40 / 1000 \times 0.0005) = 1.42 \times 10^{-5}$ mol | **1** |

| | | |
|---|---|---|
| 4(b)(ii) | $(1.42 \times 10^{-5} \times 176 \times 200 / 5) = 0.099968$<br><br>**M1** correct method to deduce mass of vit C in 5 $cm^3$ sample<br>$(1.42 \times 10^{-5} \times 176) = 2.49 \times 10^{-3}$<br>**M2** correct calculation for mass of vit C in original solution<br>$(2.49 \times 10^{-3} \times 200 / 5) = 0.099968$<br><br>**OR**<br><br>**M1** correct method to deduce amount, mol, vit C in 5 $cm^3$ sample<br>$(1.42 \times 10^{-5} \times 200 / 5) = 5.68 \times 10^{-4}$<br>**M2** correct calculation for mass of vit C in original solution = M1 $\times$ 176<br>$(5.68\ 10^{-4}) \times 176 = 0.099968$ | **2** |

# Examiner report missing
# 9701_s25_qp_21 Question 4

| | | |
|---|---|---|
| 5(a) | **M1** two ester links<br>**M2** rest of structure correct<br>  | **2** |
| 5(c)(i) | $HOOCCH_2COOH + Na_2CO_3 \rightarrow NaOOCCH_2COONa + CO_2 + H_2O$ | **1** |
| 5(e)(i) | $LiAlH_4$ | **1** |
| 5(e)(ii) | **M1** correct unambiguous organic product<br>**M2** correct balanced equation<br><br>$HOOCCH_2COOH + 8[H] \rightarrow HOCH_2CH_2CH_2OH + 2H_2O$ | **2** |

# Examiner report missing
# 9701_s25_qp_21 Question 5

| | | |
|---|---|---|
| 1(a)(i) | • (+)3 / III • –1 / I<br>• (+)5 / V • –1 / I | **2** |

| | | |
|---|---|---|
| 1(a)(iii) | ratio of moles of P : C$l$ = $\frac{30.4}{31.0}$ : $\frac{69.6}{35.5}$ = 0.98 : 1.96 (= 1 : 2)<br><br>molecular formula = $P_2Cl_4$ | **2** |

# Examiner report missing
# 9701_m25_qp_22 Question 1

| | | |
|---|---|---|
| 2(a) | electrostatic attraction<br><br>between positive metal ions / cations and delocalised electrons | **2** |

| | | |
|---|---|---|
| 2(d)(ii) | $Sr + 2H_2O \rightarrow Sr(OH)_2 + H_2$ | **1** |

# Examiner report missing
# 9701_m25_qp_22 Question 2

| | | |
|---|---|---|
| 1(a)(i) | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ | **1** |

| | | |
|---|---|---|
| 1(c) | mass of an atom of an isotope | **1** |
| | compared to the mass of the unified atomic mass unit (amu )<br>OR<br>on a scale on which a carbon-12 atom has a mass of exactly 12 units<br>OR<br>divided by $\frac{1}{12}$ mass of a carbon-12 atom | **1** |

| | | |
|---|---|---|
| 1(c)(ii) | $\frac{190.96 \times 15.18 + 192.96 \times 34.82}{50.00}$ | **1** |
| | = **192.35** | **1** |

| | | |
|---|---|---|
| 1(d)(i) | **M1** $\frac{79.5}{209.4} : \frac{20.5}{18.0} = 0.380 : 1.14 = 1:3$<br><br>OR<br><br>$18x / (102.9 + 3(35.5) + 18x) = 0.205$<br><br>$x = 3$<br><br>OR<br><br>$(102.9 + 3(35.5)) / (209.4 + 18x) = 0.795$<br><br>$x = 3$ | **1** |
| | **M2** $x = 3$ | **1** |

**Question 1**

**(a) (i)** Most candidates were able to give a correct electronic configuration of: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. A few candidates included the $3d^74s^2$ given in the question. This was not required in the answer.

**(ii)** This question was well answered with most drawings attempting to represent a sphere as a circle. A few diagrams showed the x, y and z axes drawn to represent a 3D shape.

**(iii)** Several candidates gave an answer of one, whilst many gave the correct answer.

**(b)** Most candidate answers were correct for the total number of protons and neutrons, but the total number of electron shells, for rhodium, were often incorrectly given as 4.

**(c) (i)** The definition was not well answered. Many candidates stated that it was the average mass of an isotope/weighted average mass of an isotope or the average mass of isotopes. Very few answers gave a complete definition.

**(ii)** Many candidates incorrectly used all three isotopes to calculate the $A_r$ of iridium and divided by 100 or used the two isotopes of iridium but divided by 100 instead of by 50.

**(d) (i)** Many candidates were aware that to calculate an integer value of x, the water of crystallisation in $RhCl_3 \cdot xH_2O$, a mole ratio calculation was required. The ratio of $H_2O$ to $RhCl_3$ was given/deduced as 20.5 per cent : 79.5 per cent by mass, from which a mole ratio of 0.38 : 1.14 can be calculated. Many candidates chose a slightly different route and used the expression $18x \div (209.4 + 18x) = 0.205$, to correctly calculate a vale of $x = 3$.

**(ii)** Many candidate answers mentioned that stereoisomers contained the same 'molecular formula' but omitted the fact that they also contained the same 'structural formula', which is the key feature for this definition. Several candidates correctly stated that the stereoisomers had 'different arrangements in space' but made no mention of what was arranged differently in space e.g. atoms/groups.

**(iii)** Many candidates failed to recognise that a single product would be formed and gave explanations about adding extra $CH_2$ units to the chain.

**iv)** The diagrams of the two stereoisomers of but-2-ene were often correct. Errors included confusion on the position of the C=C, in the chain, on one or both structures and incorrect groups or incorrect arrangements.

| | | |
|---|---|---|
| 2(a)(i) | chlorine = yellow-green<br>bromine = orange / brown / red<br>iodine = silver grey / black | **1** |
| 2(b)(ii) | ultraviolet | **1** |
| 2(c)(iii) | $CH_3COCH_3$ + **3**NaC*l*O → **1**CHC$l_3$ + **1**$CH_3COONa$ + **2**NaOH | **1** |

**Question 2**

**(a) (i)** The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

**(ii)** Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

**(iii)** Candidate answers that were awarded credit stated that 'white fumes' would be observed although strictly this occurs only when moisture is present.

**(iv)** Many candidate answers linked the decreasing thermal stability of hydrogen halides to weaker intermolecular forces of attraction. Some candidates only mentioned bond length and did not link the decrease in thermal stability to the decrease in bond strength of the H-Hal bond.

**(b) (i)** Many candidate answers stated that it was an 'unpaired electron or a lone pair of electrons'. The requirement for a species to be specified, that contained 'one or more unpaired electrons', was often omitted.

**(ii)** This question was well answered.

**(iii)** Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

**(c) (i)** Many candidates recognised that NaOH was required to convert chlorine to NaClO, but the requirement for the NaOH to be in aqueous solution and the conditions to be 'cold' were regularly omitted.

**(ii)** The definition of disproportionation was well answered by many candidates.

**(iii)** This equation was well known.

**(iv)** This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of $NO_2$ is also formed. Those candidates who described how the $Cl^-$ ion was formed, prior to its reaction with $Ag^+$ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form $Cl^-$. Many stated that the $Cl^-$ was contained in $CHCl_3$, and not a product from its hydrolysis.

| | | |
|---|---|---|
| 3(a)(i) | | **3** |

| element | C (graphite) | Sn |
|---|---|---|
| state and appearance | grey shiny solid | silvery solid |
| electrical conductivity | ● good/ conductor | good |
| type of bonding | ● covalent | metallic |
| type of structure | giant | ● giant |

| | | |
|---|---|---|
| 3(b)(i) | $MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$ | **1** |
| 3(e)(i) | $Si + 2Cl_2 \rightarrow SiCl_4$ | **1** |
| 3(f) | $Sn(SO_4)_2$ | **1** |

**Question 3**

**(a) (i)** This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

**(ii)** Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

**(iii)** Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

**(b) (i)** The most common errors were to give the formula for magnesium chloride as MgC$l$ and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

**(ii)** Increases was correctly stated by many candidates.

**(iii)** Many candidates gained some credit for appreciating that the ammonium carbonate donated a proton and the NaOH accepted this proton. Few answers gained full credit for stating that the ammonium ion, $NH_4^+$, is the actual species that behaves as an acid, by donating the proton, and it is the OH– ion that is accepting these protons.

**(c) (i)** Correct completion of the graph was shown by many candidates.

**(ii)** Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

**(d) (i)** There were many correct candidate answers stating an increase in the number of electrons, from $CH_4$ to $PbH_4$, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

**(ii)** The shape of a molecule of $SiH_4$ was well answered.

**(e) (i)** Many correct answers for the equation were present. A common error was to use HC$l$ as a reactant, instead of C$l_2$.

**(ii)** Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when SiC$l_4$ is added to water.

**(iii)** Credit was given to candidates for recognising that the structure and bonding of solid, $SiO_2$, was giant covalent and liquid SiC$l_4$ as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in $SiO_2$ but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid SiC$l_4$ molecules. Several answers compared the strength of the covalent bonds in both $SiO_2$ and SiC$l_4$ and many answers incorrectly stated that the structure of $SiO_2$ contained double bonds between the Si and O atoms.

**(f)** Many candidates gave the correct answer.

| | | |
|---|---|---|
| 1(a)(i) | lowest energy state owtte<br><br>examples<br>• (with) low**est** energy<br>• (with) no external energy<br>• (with) no added energy<br>• (where) electrons are not promoted to a higher energy level<br>• most energetically stable | **1** |

| | | |
|---|---|---|
| 1(b)(i) | <table><tr><th>isotope</th><th>RIM</th><th># protons</th><th># neutrons</th></tr><tr><td>$^{93}_{41}Nb$</td><td>92.91</td><td>● 41</td><td>● 52</td></tr><tr><td>$^{181}_{73}Ta$</td><td>180.95</td><td>● 73</td><td>● 108</td></tr></table> | **2** |

| | | |
|---|---|---|
| 1(b)(ii) | **M1** mass of an (atom of an) isotope | **1** |
| | **M2** relative/compared to (the mass of) the unified atomic mass unit<br>**OR**<br>on a scale in which a carbon-12 atom / isotope has a mass of exactly 12 units<br>OR<br>divided by / compared to $\frac{1}{12}$ of the mass of a carbon-12 atom / isotope<br><br>Alternative route using mass of 1 mol throughout<br>**M1** mass of one mol of an (atom of an) isotope<br>**M2** relative/compared to $\frac{1}{12}$ of the mass of 1 mol of C-12 atom / isotope<br>**OR**<br>when the mass of one mol C-12 atom / isotope is exactly 12(.000) g | **1** |

| | | |
|---|---|---|
| 1(b)(iii) | $92.91 \times 0.909 + 180.95 \times 0.091$ | **1** |
| | = 100.92 | **1** |

**Question 1**

**(a) (i)** Understanding of 'ground state' was rarely described correctly. Confusion of 'ground state' with 'natural' or 'standard' state was seen.

**(ii)** Many candidates gave the correct representation of the arrangement of electrons in the 3d and 4s orbitals.

**(iii)** The most common incorrect answer was '6'; perhaps deduced by those who considered only one of the energy levels n = 2 or n = 3.

**(b) (i)** Most candidates gave the correct deduction of the number of protons and neutrons in each species.

**(ii)** Confusion of the terms 'relative atomic mass' with 'relative isotopic mass' was seen in weaker candidate responses. Comparisons of mass were often incomplete or incorrect, examples included no identification of the specific carbon isotope.

**(iii)** Fully correct answers used the relative isotopic mass given in Table 1.1 correctly and gave their final answer to 2 decimal places. Weaker responses did not select the relevant data from Table 1.1. Final answers were not always given to the 2 decimal places specified.

| | | |
|---|---|---|
| 2(a)(i) | M1 $4Na + O_2 \rightarrow 2Na_2O$ | **1** |
| | M2 $S + O_2 \rightarrow SO_2$ | **1** |

| | | |
|---|---|---|
| 2(b)(ii) | $CH_3CN + 2H_2O + (1)H^+ \rightarrow CH_3COOH + NH_4^+$ | **1** |

**Question 2**

**(a) (i)** The formula of sodium oxide was not well known. In equations where correct formulae were shown, a significant proportion were not balanced.

**(ii)** Candidate answers were rarely given in terms of an aluminium cation and an oxide anion. When ions were described, they were generally shown with the appropriate number of outer electrons; charges on the ions were sometimes omitted or incorrect.

**(iii)** Weak candidate answers did not fully address the question and explained why the maximum oxidation state varies because of the variable number of valence electrons.

**(b) (i)** Many candidate answers referred to $Mg(OH)_2$ with a pH of a slightly acidic solution rather than weakly alkaline. The product of the reaction of $P_4O_{10}$ and $H_2O$ was not well known, $P(OH)_5$ was a common incorrect answer.

**(ii)** This equation proved to be demanding. Common errors included equations which were incorrectly balanced or showed ammonia as a product of the reaction.

**(iii)** This question was answered well by candidates. Occasionally the same structure was shown twice, usually with propan-1-ol as both answers.

**(iv)** Excellent candidate answers were seen and these explained why there were fewer $H^+(aq)$ in solution in terms of the increase in strength of O-H bond due to the positive inductive effect of the ethyl group. Inappropriate use of the term 'electronegativity' instead of 'electron density' was seen. Common misconceptions included the production of $OH^-$ ions when alcohols dissociate or that water fully dissociates or that alcohols contain $OH^-$ ions.

**(c) (i)** Many candidate answers referred to $H_2O$ although this was not relevant in **(c)(i)**. Weaker responses appreciated that there was an increase in strength of intermolecular forces but did not explain how this change arose. Many candidates gave answers in terms of an increase in size of the molecules or in terms of nuclear attraction, nuclear charge and shielding, which is incorrect. A common misconception described the increase in boiling point due to an increase in strength of permanent dipoles as the group is descended.

**(ii)** Identification of hydrogen bonds in $H_2O$ was relatively common. The relevant strength of hydrogen bonds in water compared to those intermolecular forces present in the other hydrides was not always included. Incorrect reference to $H_2S$ containing hydrogen bonds was seen and there was ambiguity in answers that referred to the relative strength of the O-H bonds rather than intermolecular hydrogen bonds.

| | | |
|---|---|---|
| 3(a)(i) | (natural = )lightning | **1** |
| | (man-made = )internal combustion engines | **1** |
| 3(a)(ii) | $2NO_2 + H_2O \rightarrow HNO_2 + HNO_3$ **OR** $4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$ | **1** |
| 3(a)(iv) | $2HNO_3 + CaO \rightarrow Ca(NO_3)_2 + H_2O$ | **1** |
| 3(a)(v) | brown fumes given off | **1** |

**Question 3**

**(a) (i)** Many candidate answers referred to lightning as responsible for one natural occurrence of $NO_2$. A common incorrect answer was acid rain. Fewer candidate answers described the main source of man-made nitrogen oxides as the internal combustion engine. Lots of candidates incorrectly identified the source as catalytic converters or 'car exhaust' as the origin of nitrogen oxides.

**(ii)** Very few candidates gave appropriate species in the equation and if they did these equations were not always balanced correctly. Many missed the significance of the word 'direct' in the question and attempted to give an equation showing the role of $NO_2$ as a catalyst in the production of sulfur-based acid rain.

**(iii)** Candidates who were familiar with this area of the syllabus described how PAN forms from $NO_2$. Common incorrect descriptions included acid rain, water, oxygen, carbon monoxide, carbon dioxide or ozone reacting with $NO_2$.

**(iv)** Correctly balanced equations were common. Typical errors included incorrect formula of nitric acid or calcium nitrate or production of $H_2$ rather than $H_2O$.

**(v)** Some candidate answers described the production of a brown gas. Others focused on the production of a white solid even though the original compound is also a white solid.

**(b) (i)** Occasionally incorrect oxidation states were seen; –1 and +3 were common incorrect answers.

**(ii)** The majority of candidates identified aluminium as the correct species that is oxidised. The weakest answers described one of the two products of the reaction.

**(iii)** Excellent answers gave a full description of the formation of a dative bond due to the lone pair on the nitrogen atom accepting a proton. Weaker candidate descriptions only referred to the ability of ammonia to form a dative bond.

**(iv)** Identification of an appropriate shape for $Al(OH)_4^-$ was present in a significant proportion of answers.

**(c) (i)** Inclusion of state symbols for ionisation energy equations are essential. Some equations showed addition of an electron, other equations were incorrectly balanced.

**(ii)** A general increase in ionisation energy was shown by most candidates. Many answers showed a larger increase in energy for one of the ionisation processes, with the exact ionisation process often incorrectly identified. Weaker answers confused the trend in ionisation energy across a period with the trend of successive ionisation energies.

**(d)** Phosphorus was sometimes identified incorrectly with ionic bonding or as an ionic structure.

**(e)** Use of the term 'simple' to identify the structure of solids made up of discrete molecules was not well known and the structure of solid nitrogen was often referred to as 'giant'.

**(f) (i)** This question highlighted many misconceptions about hybridisation, sigma and pi bonds. These include

- a triple bond involves formation of two sigma bonds and one pi bond
- triple bonds are made when sigma and pi overlap
- hybridisation in $P_2$ identified as $sp^2$ or $sp^3$
- in $P_2$ overlap of s orbitals creates a sigma bond
- formation of sigma bonds by side-on overlap of orbitals and pi bonds by head-on overlap of orbitals.

**(ii)** Many candidate responses referred to the appropriate difference in bond strength. Some answers were incomplete, with no comparison of reactivity given. Some candidates misinterpreted the question and explained why they had different bond energy values.

| | | |
|---|---|---|
| 1(a)(i) | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ | **1** |

| | | |
|---|---|---|
| 1(c) | mass of an atom of an isotope | **1** |
| | compared to the mass of the unified atomic mass unit (amu )<br>OR<br>on a scale on which a carbon-12 atom has a mass of exactly 12 units<br>OR<br>divided by $\frac{1}{12}$ mass of a carbon-12 atom | **1** |

| | | |
|---|---|---|
| 1(d)(i) | **M1** $\frac{79.5}{209.4} : \frac{20.5}{18.0} = 0.380 : 1.14 = 1:3$<br><br>OR<br><br>$18x / (102.9 + 3(35.5) + 18x) = 0.205$<br><br>$x = 3$<br><br>OR<br><br>$(102.9 + 3(35.5)) / (209.4 + 18x) = 0.795$<br><br>$x = 3$ | **1** |
| | **M2** $x = 3$ | **1** |

**Question 1**

**(a) (i)** Most candidates were able to give a correct electronic configuration of: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. A few candidates included the $3d^74s^2$ given in the question. This was not required in the answer.

**(ii)** This question was well answered with most drawings attempting to represent a sphere as a circle. A few diagrams showed the x, y and z axes drawn to represent a 3D shape.

**(iii)** Several candidates gave an answer of one, whilst many gave the correct answer.

**(b)** Most candidate answers were correct for the total number of protons and neutrons, but the total number of electron shells, for rhodium, were often incorrectly given as 4.

**(c) (i)** The definition was not well answered. Many candidates stated that it was the average mass of an isotope/weighted average mass of an isotope or the average mass of isotopes. Very few answers gave a complete definition.

**(ii)** Many candidates incorrectly used all three isotopes to calculate the $A_r$ of iridium and divided by 100 or used the two isotopes of iridium but divided by 100 instead of by 50.

**(d) (i)** Many candidates were aware that to calculate an integer value of x, the water of crystallisation in $RhCl_3 \cdot xH_2O$, a mole ratio calculation was required. The ratio of $H_2O$ to $RhCl_3$ was given/deduced as 20.5 per cent : 79.5 per cent by mass, from which a mole ratio of 0.38 : 1.14 can be calculated. Many candidates chose a slightly different route and used the expression $18x \div (209.4 + 18x) = 0.205$, to correctly calculate a vale of $x = 3$.

**(ii)** Many candidate answers mentioned that stereoisomers contained the same 'molecular formula' but omitted the fact that they also contained the same 'structural formula', which is the key feature for this definition. Several candidates correctly stated that the stereoisomers had 'different arrangements in space' but made no mention of what was arranged differently in space e.g. atoms/groups.

**(iii)** Many candidates failed to recognise that a single product would be formed and gave explanations about adding extra $CH_2$ units to the chain.

**iv)** The diagrams of the two stereoisomers of but-2-ene were often correct. Errors included confusion on the position of the C=C, in the chain, on one or both structures and incorrect groups or incorrect arrangements.

| | | |
|---|---|---|
| 2(a)(i) | chlorine = yellow-green<br>bromine = orange / brown / red<br>iodine = silver grey / black | **1** |
| 2(a)(iii) | green colour disappears | **1** |
| 2(b)(ii) | ultraviolet | **1** |
| 2(b)(iii) | $C_2H_6 + Cl^{\bullet} \rightarrow C_2H_5^{\bullet} + HCl$ | **1** |
| | $C_2H_5^{\bullet} + Cl_2 \rightarrow C_2H_5Cl + Cl^{\bullet}$ | **1** |
| 2(c)(iii) | $CH_3COCH_3 + \mathbf{3}NaClO \rightarrow \mathbf{1}CHCl_3 + \mathbf{1}CH_3COONa + \mathbf{2}NaOH$ | **1** |

**Question 2**

**(a) (i)** The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

**(ii)** Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

**(iii)** Candidate answers that were awarded credit stated that 'white fumes' would be observed although strictly this occurs only when moisture is present.

**(iv)** Many candidate answers linked the decreasing thermal stability of hydrogen halides to weaker intermolecular forces of attraction. Some candidates only mentioned bond length and did not link the decrease in thermal stability to the decrease in bond strength of the H-Hal bond.

**(b) (i)** Many candidate answers stated that it was an 'unpaired electron or a lone pair of electrons'. The requirement for a species to be specified, that contained 'one or more unpaired electrons', was often omitted.

**(ii)** This question was well answered.

**(iii)** Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

**(c) (i)** Many candidates recognised that NaOH was required to convert chlorine to NaClO, but the requirement for the NaOH to be in aqueous solution and the conditions to be 'cold' were regularly omitted.

**(ii)** The definition of disproportionation was well answered by many candidates.

**(iii)** This equation was well known.

**(iv)** This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of $NO_2$ is also formed. Those candidates who described how the $Cl^-$ ion was formed, prior to its reaction with $Ag^+$ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form $Cl^-$. Many stated that the $Cl^-$ was contained in $CHCl_3$, and not a product from its hydrolysis.

| | | |
|---|---|---|
| 3(a)(i) | <table><tr><td>element</td><td>C (graphite)</td><td>Sn</td></tr><tr><td>state and appearance</td><td>grey shiny solid</td><td>silvery solid</td></tr><tr><td>electrical conductivity</td><td>● good/ conductor</td><td>good</td></tr><tr><td>type of bonding</td><td>● covalent</td><td>metallic</td></tr><tr><td>type of structure</td><td>giant</td><td>● giant</td></tr></table> | **3** |
| 3(e)(i) | $Si + 2Cl_2 \rightarrow SiCl_4$ | **1** |
| 3(f) | $Sn(SO_4)_2$ | **1** |

**Question 3**

**(a) (i)** This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

**(ii)** Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

**(iii)** Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

**(b) (i)** The most common errors were to give the formula for magnesium chloride as $MgCl$ and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

**(ii)** Increases was correctly stated by many candidates.

**(iii)** Many candidates gained some credit for appreciating that the ammonium carbonate donated a proton and the NaOH accepted this proton. Few answers gained full credit for stating that the ammonium ion, $NH_4^+$, is the actual species that behaves as an acid, by donating the proton, and it is the OH– ion that is accepting these protons.

**(c) (i)** Correct completion of the graph was shown by many candidates.

**(ii)** Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

**(d) (i)** There were many correct candidate answers stating an increase in the number of electrons, from $CH_4$ to $PbH_4$, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

**(ii)** The shape of a molecule of $SiH_4$ was well answered.

**(e) (i)** Many correct answers for the equation were present. A common error was to use $HCl$ as a reactant, instead of $Cl_2$.

**(ii)** Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when $SiCl_4$ is added to water.

**(iii)** Credit was given to candidates for recognising that the structure and bonding of solid, $SiO_2$, was giant covalent and liquid $SiCl_4$ as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in $SiO_2$ but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid $SiCl_4$ molecules. Several answers compared the strength of the covalent bonds in both $SiO_2$ and $SiCl_4$ and many answers incorrectly stated that the structure of $SiO_2$ contained double bonds between the Si and O atoms.

**(f)** Many candidates gave the correct answer.

| | | |
|---|---|---|
| 1(a)(i) | **M1** ($N_2$ molecules /molecule(s) of $N_2$) have a strong triple (covalent) bond<br>**M2** ($N_2$ molecules /molecule(s) of $N_2$) are non-polar | **2** |

| | | |
|---|---|---|
| 1(c)(i) | both states shown as solid | **1** |

| | | |
|---|---|---|
| 1(c)(ii) | **M1** *correctly balanced equation for aluminium oxide*<br>$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$<br><br>**M2** *correctly balanced equation for $P_4O_{10}$ and an excess of NaOH*<br>$P_4O_{10} + 12NaOH \rightarrow 4Na_3PO_4 + 6H_2O$ | **2** |

**Question 1**

This question assessed inorganic chemistry, with some aspects of physical chemistry. This tested principally in the AO1 category. Candidates showed some facility of recall but were equally divided in response when asked to apply their knowledge to unfamiliar circumstances.

**(a)(i)** This standard recall question could be answered with easy reference to syllabus points.

**(ii)** With reference to the syllabus, orbital overlap can be directly quoted; knowledge of the bonding in $N_2$ is also straightforward, so the number of $\sigma$ and $\pi$ bonds should be readily accessible.

**(b)(i)** Many candidates answered this question well.

**(ii)** This question gave rise to many variations of answer. Candidates are advised to be brief with their answers: often, correct responses were contradicted by (incorrect) explanations of valence electrons.

**(c)(i)** Many candidates answered this question well.

**(ii)** The reaction of $P_4O_{10}$ was better known than that of $Al_2O_3$. The latter is not less common but requires better honed balancing skills if constructed rather than written in recall.

**(d)(i)** Many candidates answered this question well.

**(ii)** Many candidates were able to identify the type of reaction that occurs.

| | | |
|---|---|---|
| 2(a) | 58% / no change / stays the same **AND** catalysts don't affect yield / iron is a catalyst | **1** |

| | | |
|---|---|---|
| 2(d)(i) | **M1** catalyst<br><br>**M2** *2 equations to show the catalytic activity of $NO_2$ in formation of $SO_3$ from $SO_2$*<br>• $SO_2 + NO_2 \rightarrow NO + SO_3$<br>• $2NO + O_2 \rightarrow 2NO_2$<br><br>**M3** *describe production of $H_2SO_4$ from $SO_3$ and water (can be shown in an equation).*<br>$SO_3$ then reacts with (rain)water / moisture / water vapour to make sulfuric acid / $H_2SO_4$ OR $SO_3 + H_2O \rightarrow H_2SO_4$ | **3** |

**Question 2**

Although **(d)** focused on inorganic chemistry, the majority of this question was based on equilibria and rates. Better performing candidates were able to discern the differing aspects here: rates and collision theory formed one notion; equilibria, yield and Le Chatelier's principle the other.

**(a)** This was generally well answered, with many candidates identifying the null effect of a catalyst on equilibrium position. The best answers stated this explicitly.

**(b)(i)** This was well answered by candidates, who could link lower yield with the exothermic reaction. Precision of wording was key here.

**(ii)** Many candidates attempted to answer this rates question in terms of equilibrium yield and were unable to gain full credit as a result. If an item pairs rate and yield within a question, candidates are advised to read the stem of the question carefully and to give relevant answers.

**(c)(i)** Many candidates were unable sufficiently to distinguish between $K_c$ and $K_p$. Square brackets must not be used in $K_p$ expressions to avoid confusion with concentration variables. Partial pressure variables can be given as $p_X$, $P_X$, $p(X)$, $P(X)$ or $pp(X)$.

**(ii)** Many numerical answers were correct, though common errors included not adjusting the equilibrium concentrations of $N_2$ and $H_2$, or not accounting for **2**$NH_3$ or **3**$H_2$ in the stoichiometric equation.

**(iii)** This was generally well answered by candidates, although some errors of powers of ten were seen in a minority of responses.

**(d)(i)** Candidates' knowledge of this nitrogen chemistry seemed largely secure, though correct quotation of three equations (two showing $NO_2$ catalysis and one of the conversion of $SO_3$ to acid rain) were needed to gain full credit, alongside a statement of the role of $NO_x$.

**(ii)** There were many good partial answers to this question, with most referring to PAN but with less precision on how this is produced.

| | | |
|---|---|---|
| 3(a) | $H_2(g) + ½O_2(g) \rightarrow H_2O(l)$<br>**M1** correct equation only<br>**M2** correct state symbols based on this equation | **2** |

| | | |
|---|---|---|
| 3(c)(i) | giant (molecular and) covalent | **1** |

| | | |
|---|---|---|
| 3(c)(ii) | **M1** *give correct expression or calculate mol B compared to mol C*<br>78.26 / 10.8 (= 7.246 / 7.25 mol B) **AND** 21.74 / 12 (=1.81(2) mol C)<br><br>**M2** Find B : C ratio **AND** use this to state empirical formula<br>7.246 to 7.25 / 1.817 to 1.82 = 3.99 to 3.98 mol B : 1 mol C<br>(so) $B_4C$ OR $CB_4$ | **2** |

**Question 3**

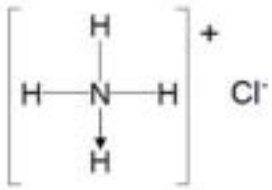
Candidates are urged to look carefully at the information provided in each item: here, this proved helpful and key to answering the ensuing physical chemistry questions.

**(a)** This was well answered by candidates. Standard states were required, that of water being (l).

**(b)** Common errors included incorrect stoichiometry or reverse parity for the answer. On the whole, the principle of Hess's Law seemed to be understood where answered.

**(c)(i)** There were very many different answers here; it seemed that candidates were supposing that the reaction of 'metallic' boron and non-metallic carbon must result in a giant ionic lattice.

**(ii)** This was well answered by candidates who correctly used the $A_r$ values of B and C. Some confusion was evident over the method of deciding the ratio of B : C. Answers without working were not credited, nor were answers trying to reason a formula from the ratio of putative charges of ions.

| | | |
|---|---|---|
| 4(a) |  <br><br>**OR**<br><br>Correct representation or label showing<br>**M1** (ionic bond between) $NH_4^+$ and $Cl^-$ **AND** 4 N—H covalent bonds OR 4 shared pairs of electrons in $NH_4$<br><br>**M2** 1 N—H coordinate bond OR arrow on bond/line in correct direction or only 1 bonding pair using the same symbol for electrons1 | **2** |
| 4(b)(i) | acid–base reaction | **1** |
| 4(b)(ii) | $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$ | **1** |

**Question 4**

Many individual parts of this question were answered well. Parts **(a)**, **(c)(i)**, **(e)(ii)** and **(f)(ii)** were clearly the most challenging, often as candidates were required to apply several independent parts of their knowledge of the syllabus.

**(a)** Candidates were not strictly required to give dot-and-cross diagrams, though these and modified Lewis diagrams often produced the clearest responses. Many answers incorrectly showed a pentavalent nitrogen atom — a kind of $NH_5$ species, with little understanding of the $NH_4^+$ $Cl^-$ superstructure. The dative bond needed to be clearly distinct from other covalent bonds and ionic charges needed to be given.

**(b)(i)** There were many correct answers to this question.

**(ii)** Different versions of this answer were offered, but a large number were a simple addition of $NH_4^+$ and $OH^-$, which did not gain credit.

**(c)(i)** This was well answered by candidates whose knowledge extended to terminology associated with electron-pair donors.

**(ii)** This was found to be the most challenging item on the paper which required candidates to assess either the basicity or nucleophilicity of each species. Opposing answers were therefore acceptable with enough reasoning.

**(d)** This was generally well answered by candidates, but the single-step nature of the $S_N2$ mechanism needed to be clearly evidenced by one transition state; the slower rate of this reaction by a higher energy of the TS relative to either produced in $S_N1$. Candidates frequently accessed the former part but often then erroneously showed some kind of catalysed reaction with a lower activation energy.

**(e)(i)** This question was not generally well answered, as candidates were not able to associate electron movement with the correct stage of the reaction and often were very imprecise with their positioning of curly arrows. Many answers had the arrowhead and tail reversed, with a lot of responses suggesting electron movement *from* extra $H^+$ species.

**(ii)** This question was reasonably well answered.

**(iii)** This question was generally well answered, although stray vowels or unspecific/incorrect numbering were observed. Candidates are reminded that the longest carbon chain is the important root of a molecule's name.

**(f) (i)** Candidates were able to provide a correct answer here by analogy of the 2-bromobutylium species on the left-hand side.

**(ii)** This was found to be a demanding question, although candidates who recognised the necessary reference to the stability of the intermediate were able to give a reasonably straightforward AO1 account for full credit.

| | | |
|---|---|---|
| 1(a) | <table><tr><td>$Mg^{2+}$</td><td>24</td><td>12</td><td>10</td></tr><tr><td>$Al^{3+}$</td><td>27</td><td>13</td><td>10</td></tr></table><br>4 correct = 2 marks<br>2 or 3 correct = 1 mark<br>0 or 1 correct = 0 marks | **2** |
| 1(e)(iii) | $P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$ | **1** |
| 1(f)(i) | reacts with both acids and bases **OR** shows both acidic and basic behaviour | **1** |
| 1(f)(ii) | $Al(OH)_3 + NaOH \rightarrow NaAl(OH)_4$ | **1** |

**Question 1**

**(a)** The majority of answers were correct.

**(b)** Some excellent explanations were seen. Occasionally the aluminium ion was incorrectly described as larger than the magnesium ion. In general, the explanations focussed, at least in part, on the difference in nuclear attraction which arises as a result of the difference in nuclear charge and the same shielding effect of both species.

**(c)** Clearly labelled diagrams, which showed a regular arrangement of sodium cations surrounded by delocalised electrons, were seen in a significant proportion of answers. Poorer diagrams showed randomly placed positive ions or made no reference to 'delocalised' electrons. Occasionally, the positive symbols were described incorrectly as 'nuclei' or 'protons' or 'atoms'.

**(d) (i)** The majority identified the structure and bonding present in silicon. A much smaller proportion used this knowledge to explain why silicon had a high melting point in terms of the high energy requirement required to break many strong covalent bonds. A common misconception described intermolecular forces in a giant molecular structure.

**(ii)** Of those that annotated Fig.1.1, many correctly compared the melting points of P, S and C*l* with those of the three metals provided. Comparison of the melting points between P, S and C*l* was less well known.

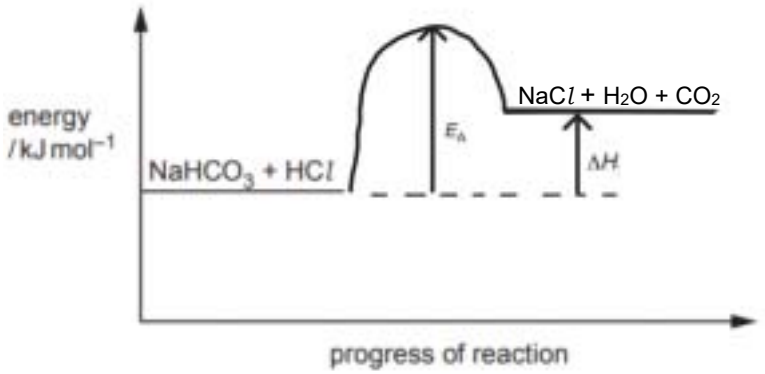
**(e) (i)** This question proved to be demanding. The state of both sodium oxide and sulfur dioxide under room conditions was not well known. Misunderstanding of the term 'state of oxide' and oxidation number was seen on occasions. A common misconception described aqueous solutions of sodium oxide with a pH of 7.

**(ii)** Correct identification of this reaction as acid–base or neutralisation was common.

**(iii)** The product for the reaction of $P_4O_{10}$ with an excess of water was well known. Equations were not always balanced.

**(f) (i)** The majority of answers demonstrated a good understanding of the term amphoteric.

**(ii)** There was some confusion with this reaction and the reaction of aluminium oxide and sodium hydroxide. The correct formula of the product was not well known.

| | | |
|---|---|---|
| 2(a) |  <br>*All three correct (2 marks) or two correct (1 mark):*<br>• profile for endothermic reaction (include 'hump')<br>• arrow representing $E_A$<br>• arrow representing $\Delta H$ **OR** (+) 27.2 | **2** |
| 2(b)(i) | no more fizzing / effervescence stops | **1** |
| 2(b)(ii) | **M1** calculate Q from = 50 × 4.18 × (26.2 – 19.6)) = 1379.(4)<br>**M2** *Use M1 to find energy change when 1 mol $Na_2CO_3$ / 2 mol HCl reacts*<br>**M1** ÷ 0.04 = 34485<br>**M3** *express Q as $\Delta H_{(1)}$ in kJ mol$^{-1}$* = – (value for **M2**) ÷ 1000<br>= –34.5 kJ mol$^{-1}$<br><br>*Alternative method*<br>**M1** *evidence of use Q = $mc\Delta T$* = 50 × 4.18 / 1000 × (26.2 – 19.6) = 1.3794<br>**M2** *Use M1 to find energy change when 1 mol $Na_2CO_3$ / 2 mol HCl reacts*<br>**M1** ÷ 0.04 = 34.485<br>**M3** *express Q as $\Delta H_{(1)}$ in kJ mol$^{-1}$* = – (value for **M2**) *(expressed in kJ)*<br>= –34.5 kJ mol$^{-1}$ | **3** |
| 2(c) | $Ca(NO_3)_2 \rightarrow CaO + 2NO_2 + ½O_2$<br><br>**M1** correct formula for calcium nitrate: $Ca(NO_3)_2$<br>**M2** correct balanced equation | **2** |

**Question 2**

**(a)** Appropriate profiles for an endothermic reaction were common. In general, representation of activation energy and enthalpy change were correct. Some answers showed very imprecise location of arrows on the diagram or showed arrows which did not point in the correct direction.

**(b) (i)** Good answers described an appropriate observation based on the idea that no more gas is produced when the reaction stops.

**(ii)** This calculation proved to be demanding. The specific heat capacity of water was not always given to the appropriate number of significant figures. The step to convert the amount of energy produced in the experiment to the amount of energy produced per mol was often confused. A significant proportion of answers included a negative sign for the enthalpy change value to reflect the exothermic nature of the reaction. Some answers were given in terms of J $mol^{-1}$ rather than kJ $mol^{-1}$, as directed in the question.

**(iii)** The best answers showed a correct Hess' cycle in their working. Weaker responses tended to use appropriate stoichiometry in their calculation but confused the algebraic relationship between $\Delta H_1$ and $\Delta H_r$ in the calculation. A relatively high proportion of candidates did not attempt to answer this question.

**(c)** Many answers deduced that **Z** was a Group 2 nitrate. Identification of calcium as a Period 4 element presented some difficulties and $Sr(NO_3)_2$ was seen on occasions. Production of the balanced equation for this thermal decomposition proved to be demanding.
The weakest responses identified the brown gas as bromine and gave equations of Group 2 bromides producing bromine.

**Question 5**

**(a) (i)** Those candidates with a good understanding of mass spectrometry realised that the peaks shown in Table 5.1 represented the molecular ion peak, $M^+$, and its $(M+1)^+$ peak and used the appropriate formula to calculate the relative abundance of the M+1 peak. Some weaker responses based their calculation on an incorrect formula.

**(ii)** The strongest responses engaged with the information about compound **W**. Weaker responses did not engage fully with the question details and answered in terms of structural formulae. Some formulae gave the fragments with no charges or negative charges.

**(b) (i)** An understanding that an O–H bond, due to a carboxyl functional group, produces a broad stretch within the range 2500–3000 $cm^{-1}$ was seen in the best responses. The weakest responses did not engage with the details given for compound **W** and its reaction to produce **X**, and listed every potential functional group which might account for bond **A** and bond **B**, including amide and ester functional groups.

**(ii)** Of those candidates that correctly identified **B** as a C=O of a carbonyl group, some candidates ignored the reaction conditions stated in the question and represented **X** as an aldehyde rather than a ketone.

**(c) (i)** A significant proportion of candidates correctly named the functional group present in **Y**, based on the information provided.

**(ii)** Many equations were shown with the correct formula of the organic product; not all equations were balanced. Identification of $H_2O$ rather than $H_2$ as the other product was seen.

**(iii)** Those candidates who applied knowledge of the behaviour of different types of alcohol and carbonyl compounds correctly deduced the structure of **Y**. Confusion about the chemical reactions of alcohols and carbonyl compounds was seen in weaker responses.

| | | |
|---|---|---|
| 1(a) | <table><tr><td>halogen</td><td>colour at 293 K</td></tr><tr><td>chlorine</td><td>green</td></tr><tr><td>bromine</td><td>brown</td></tr><tr><td>iodine</td><td>(dark) grey</td></tr></table><br>All three correct for one mark | **1** |
| 1(c)(i) | $Br_2 + 2I^- \rightarrow 2Br^- + I_2$ | **1** |
| 1(d)(i) | $NaBr + H_2SO_4 \rightarrow HBr + NaHSO_4$<br>**OR**<br>$2NaBr + H_2SO_4 \rightarrow 2HBr + Na_2SO_4$ | **1** |

**Question 1**

**(a)** The colour of the halogens (at 293 K) was not often answered correctly by candidates. Common errors included chlorine described as yellow and iodine as a purple gas. Several candidates gave the colours of silver halides.

**(b)** The volatility of the halogens chlorine, bromine and iodine is associated specifically with the increased strength of the instantaneous dipole-induced dipole forces between the molecules due to the increase in the number of electrons in these molecules. Many candidates correctly stated that the volatility decreases from chlorine to bromine to iodine. This trend was then incorrectly attributed or linked to the electronegativity variations of the elements or the attraction between the nucleus and the outer electrons.

**(c) (i)** The majority of candidates used the full formula equation rather than ionic for the reaction of bromine with sodium iodide. Many answers included the spectator ion $Na^+$ in the equation.

**(ii)** It was recognised by several candidates that bromine, when reacted with sodium iodide, behaved as an oxidising agent to form iodine and a bromide. Many explanations then incorrectly referred to a change in oxidation number, or movement of electrons, in iodine instead of the iodide ions.

**(d) (i)** Many candidates gave a correct equation for the reaction of NaBr with concentrated sulfuric acid. Common errors were $NaSO_4$, for sodium sulfate(VI).

**(ii)** Candidates often identified NaC$l$ as the salt that produced the largest percentage of hydrogen halide. The explanations frequently referred to the reactivity of the sodium halides instead of describing the trend in the hydrogen halides or halide ions as reducing agents.

The most common incorrect answers identified ‘sodium iodide’ producing the largest percentage yield of hydrogen halide as ‘it is the most reactive’.

| | | |
|---|---|---|
| 2(a)(i) | hydrolysis | **1** |

| | | |
|---|---|---|
| 2(c)(i) | <table><tr><td></td><td>oxidation number in $Mg_3N_2$</td></tr><tr><td>magnesium</td><td>(+)II / 2(+)</td></tr><tr><td>nitrogen</td><td>–III/ 3 –</td></tr></table> Both correct for one mark | **1** |

| | | |
|---|---|---|
| 2(c)(ii) | $Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$ | **1** |

| | | |
|---|---|---|
| 2(d)(i) | BN | **1** |

**Question 2**

**(a) (i)** This question was well answered.

**(ii)** The majority of answers gave a formula for the different Period 3 element when the question asked for a name. Common errors included phosphorus chloride, omitting the oxidation number, and phosphorus pentachloride, which is a solid.

**(b) (i)** The dot-and-cross diagram of $SCl_2$ was well answered. A common error included annotating the electrons around the S atom as 'x' in contradiction of the rubric.

**(ii)** The VSEPR theory was not well known when applied to a central S atom with two bonding pairs and two lone pairs of electrons in the outer shell. The majority of candidates focused on the two bonding pairs in $SCl_2$, stating that the shape was linear with a bond angle of 180°.

**(c) (i)** This question was well answered.

**(ii)** Most candidates gave a correct equation for the reaction of magnesium nitride with water. The most common error was an incorrect formula for magnesium nitride and magnesium hydroxide given as MgOH or MgO.

**(iii)** The marks were awarded to a minority of candidates. Very few candidates stated that hydroxide ions were present either from the dissociation of $Mg(OH)_2$ in solution, or from a reaction between ammonia and water.

**(d) (i)** The empirical formula of boron nitride was well answered with a common error of Br used to represent boron.

**(ii)** Many candidates gave the correct answer of graphite that has a structure similar to that of boron nitride. $SiO_2$ and graphene were common incorrect answers.

| | | |
|---|---|---|
| 3(a) | **M1** (if a change in conditions occurs) the equilibrium shifts<br>**M2** to minimise the change in conditions | **2** |

| | | |
|---|---|---|
| 3(c)(i) | **M1** *deduce the concentration of $SCN^-$ present in initial mixture*<br>[$SCN^-$ eq'm] = [$Fe^{3+}$ eq'm] = [original $SCN^-$] – [$FeSCN^{2+}$ produced]<br>$1.30 \times 10^{-3} + 0.300 \times 10^{-3} = 1.60 \times 10^{-3}$ mol $dm^{-3}$<br><br>**M2** *find initial amount of $Fe^{3+}$ initially*<br>M1 $\times$ 25 / 1000 = amount of $Fe^{3+}$ added initially = $4.00 \times 10^{-5}$ (mol) | **2** |

**Question 3**

**(a)** Defining Le Chatelier's principle proved challenging for some candidates with a common answer being a definition of a dynamic equilibrium, in terms of the rates of forward and backward reactions. A further error referred to a shifting of the system rather than of the equilibrium, following a change in conditions.

**(b)** Many candidates were able to deduce the directed changes to an equilibrium subjected to a change in temperature. The most common incorrect answer was 'no change' for the effect of temperature change on the numerical value of $K_c$, the equilibrium constant.

**(c) (i)** A minority of candidates were able to calculate the initial moles of $Fe^{3+}(aq)$ in an equilibrium mixture: $Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$, where the concentrations of two species in the equilibrium mixture were provided in the question.

Some candidates correctly calculated the moles of $SCN^{-}(aq)$ and $FeSCN^{2+}(aq)$ at equilibrium, but then proceeded to subtract these two values instead of adding them, to give the initial moles of $Fe^{3+}(aq)$.

**(ii)** Many candidates were unable to calculate a correct value for the $K_c$, for this reaction, but gained credit for the correct units.

| | | |
|---|---|---|
| 4(a) | **M1** (enthalpy change / energy change) when one mole of a compound / substance is formed<br>**M2** from its elements in their standard states | **2** |

| | | | |
|---|---|---|---|
| 4(b)(i) | | equation for reaction | $\Delta H_{formation}$ | **2** |
| | $Fe_2O_3$ | $2Fe\,(s) + \frac{3}{2}O_2\,(g) \rightarrow Fe_2O_3\,(s)$ | –824.2 | |
| | CO | $C\,(s) + \frac{1}{2}O_2\,(g) \rightarrow CO\,(g)$ | –110.5 | |
| | $CO_2$ | $C\,(s) + O_2\,(g) \rightarrow CO_2\,(g)$ | –393.5 | |

**Question 4**

**(a)** The majority of candidates recognised that the definition of enthalpy change of formation involved an enthalpy change for the formation of one mole of compound from its elements. Most answers omitted that the elements were required to be in their 'standard states'.

**(b) (i)** The award of full credit was rare. Many equations omitted the state symbols and/or gave correctly balanced equations for two moles of product. It was common to see $Fe^{3+}$, for Fe, and O for $O_2$.

**(ii)** The calculation of the enthalpy change for the reaction: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ proved challenging for many candidates. The common errors included:

- use of incorrect stoichiometry
- use of an enthalpy value for the formation of Fe and
- incorrect use of enthalpy cycles.

| | | |
|---|---|---|
| 6(a)(i) | (molecules with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space | **1** |

| | | |
|---|---|---|
| 6(c)(i) | **M1** deduce n = 12 (from **Y** – 1C (in methanol))<br>**M2** $(100 \times x) \div (1.1 \times 100) = 12$ (so) $x = 13.2$ | **2** |

| | | |
|---|---|---|
| 6(c)(ii) | <table><tr><th></th><th>observation on addition of 2,4–DNPH</th></tr><tr><td>**Y**</td><td>orange precipitate</td></tr><tr><td>**Q**</td><td>no precipitate</td></tr></table><br>Both correct for one mark | **1** |

| | | |
|---|---|---|
| 6(c)(iii) | $ROH + Na \rightarrow RONa + 1/2H_2$<br>(0.002 mol **Q** produced 0.002 mol $H_2$ gas so) 2 OH groups<br>**M1** *answer indicates that OH group(s) in **Q** react with Na to produce the $H_2$ in the ratio 1 mol OH : ½ mol $H_2$*<br>**M2** *uses data to show 2OH groups* | **2** |

**Question 6**

**(a) (i)** The definition of stereoisomerism was one of the least well answered questions. The most common error was to state that 'molecules had the same molecular formula but different structural formula'. Few mentioned that it was the arrangement of the atoms or groups in space.

**(ii)** Very few answers gained full credit here. Some candidates identified a C=C double bond and two chiral centres and then incorrectly stated that this gave (2 × 3 =) 6 stereoisomers instead of using the correct expression $2^3$ (= 8).

**(iii)** The molecular formula of **V** was answered quite well. There were no common errors.

**(iv)** Many candidates correctly identified an alkene and carbonyl group. The most common error was to omit the ester group and substitute it for an alcohol group or a carboxyl/carboxylic acid group.

**(b) (i)** The question required an identification of the role of reagent **T**. Candidates who stated that **T** behaved as a reducing agent gained full credit. Those candidates who ventured to explain the behaviour of **T** with each functional group generally understood that it reduced the carbonyl group in **V** to an alcohol group in **W**. The reaction of the C=C to from a saturated chain attracted a variety of incorrect answers ranging from acting as a catalyst to hydrolysing the alkene group.

**(ii)** This question was well answered.

**(c) (i)** Candidates found this very challenging. Many answers used the m/e values of 200 and 201, from the mass spectrum, instead of the relative abundance values of 100 and x, and attempted to average these values to determine the relative atomic mass.

Creditworthy answers deduced that **Q** contained 12 C atoms and then used this value of *n* in the expression $n = \dfrac{100 \times \text{abundance of M} + 1 \text{ ion}}{1.1 \times \text{abundance of M} + \text{ion}}$

to calculate the abundance of the (M + 1) ion which is equivalent to the value of x.

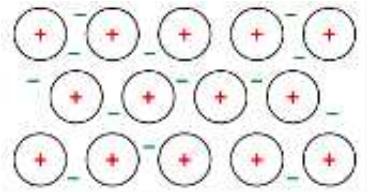
**(ii)** This question was answered correctly by many candidates with the most common error being to omit 'precipitate' with the orange colour for the observation with 2,4-DNPH with **Y**.

**(iii)** The majority of candidates did not recognise that the volume of hydrogen gas was collected at s.t.p., in the question, which, from the Table of data, is 22.4 $dm^3 mol^{-1}$.

Various methods were used to calculate the moles of gas produced to include the ideal gas equation and 24.0 $dm^3 mol^{-1}$.

Those candidates who managed to calculate the number of moles of gas then needed to appreciate that the ratio of hydroxyl groups to hydrogen gas was 2 : 1. The award of full credit was rare.

**(iv)** Many candidates gave correct absorption ranges for one of the two carbonyl absorptions in **Y**, for the ketone and ester functional groups. Several answers gave a range of absorptions for a carboxyl group in **Q** as a difference in the infrared spectrum, having not appreciated that **Q** contains **only** hydroxyl groups, mentioned in the stem of **(c)(ii)**.

| | | |
|---|---|---|
| 1(a) |  <br>labelled diagram shows:<br>• regular arrangement of positive ions<br>• **AND** surrounded by / (sea of) delocalised electrons | **1** |

| | | |
|---|---|---|
| 1(b)(i) | $2Bi + 3H_2O \rightarrow Bi_2O_3 + 3H_2$ | **1** |

| | | |
|---|---|---|
| 1(e) | moles of $NaBiO_3$ = 21.50 / 1000 × 0.100 (= $2.15 \times 10^{-3}$ mol) | **1** |
| | moles of $Mn^{2+}$ in dil. = $2.15 \times 10^{-3} \times 2 / 5 \times 40$ (= 0.0344 mol) | **1** |
| | conc. saturated $Mn^{2+}$ = 0.0344 × 1000 / 100 **= 0.344 mol $dm^{-3}$** | **1** |

**Question 1**

This was a broad-based question, encompassing ideas of physical, inorganic and organic chemistry. Candidates were adept with simple definitions, but application of ideas showed some deficit of understanding.

**(a)** Many diagrams were well presented, but a substantial number neglected to label the particles: mention of delocalisation of electrons was crucial. A sizable minority incorrectly identified the positive particles as protons or nuclei.

**(b) (i)** This question was mostly well answered, though errors of balancing and '$Bi_2$' were often seen.

**(ii)** Although many candidates secured full credit here, common errors were not to link deduction and reasoning (e.g. giant *because* high melting point) and to offer contradictory statements (e.g. giant molecular and ionic bonding). In many cases, brevity served better. Some candidates appeared confused about the compound they were being asked to describe, $Bi_2O_3$, rather than bismuth itself.

**(c) (i)** This was answered well, though answers of +6 were commonly seen with respect to bismuth.

**(ii)** Many different answers were offered here, including $NaBH_4$ and $O_2$, whereas candidates needed to refer to a compound given in the equation.

**(d) (i)** This was generally well answered. It is preferable to refer to transfer of electrons when explaining the term oxidising agent.

**(ii)** Some contradictions were observed here, such as 'cold dilute … heat under reflux.'

**(iii)** Many candidates answered this well. Common errors included a switch of answers in the case of Tollens' reagent and neglecting to mention a precipitate or solid in the iodoform and DNPH reactions.

**(iv)** Many candidates tried unsuccessfully to include $NaBH_4$ in their answers. Candidates are reminded that even with the use of [H], there is a need for a stoichiometric balancing number. Use of $CH_3COH$ is discouraged as a method of representing aldehydes.

**(e)** The initial stages of most calculations were good, but many candidates then forgot to multiply by 40 (to scale up to 1 $dm^3$ from 25 $cm^3$) or to multiply the concentration by 10 (1000 ÷ 100). In general, working needed to be shown more clearly, as some inventive methods could otherwise have given a seemingly correct answer.

| | | |
|---|---|---|
| 2(a) | <table><tr><td>formula of chloride</td><td>● NaCl</td><td>● $MgCl_2$</td><td>● $SiCl_4$</td></tr><tr><td>structure</td><td>giant</td><td>● giant</td><td>● simple</td></tr><tr><td>bonding of chloride</td><td>● ionic</td><td>● ionic</td><td>covalent</td></tr><tr><td>pH of solution formed</td><td>● 7</td><td>6.2</td><td>● 1–4</td></tr></table> | **3** |
| 2(b)(i) | disproportionation | **1** |
| 2(b)(iii) | $3Cl_2 + 6NaOH \rightarrow NaClO_3 + 5NaCl + 3H_2O$ | **1** |

**Question 2**

This was a holistic question based around the chemistry of chlorine and chlorides. Candidates are recommended to give brief answers to one-mark questions and to draw diagrams clearly.

**(a)** Common errors were found in the formulae of the chlorides, the structure of $SiCl_4$ and in the pH of a $NaCl$ solution (neutral so pH 7; candidates often seemed to think that: $NaCl + H_2O \rightarrow HCl + NaOH$).

**(b) (i)** Many variant and phonetic spellings of 'disproportionation' were observed.

**(ii)** There was an issue with this question in that water was not included as one of the products of the reaction, so full marks were awarded to all candidates to make sure no candidates were disadvantaged. This has been corrected in the published version of the paper.

**(iii)** This equation was rarely constructed correctly: where more valid attempts were made, common errors were either that $NaClO$ was given as the oxidation product or that the correct stoichiometric coefficients were omitted.

**(iv)** This was answered well by most candidates, although a number offered effervescence as an observation.

**(c) (i)** Octahedral was the most common answer for $PCl_6^-$, although commonly hexagonal or octagonal were seen (and not credited). A broad range of answers were given for $PCl_4^+$, with 'trigonal pyramidal' (missing the bi- prefix) and 'bipyramidal' (missing the base shape) among them.

**(ii)** **J** was often correctly identified as $H_2O$/water. Hydration or displacement were often given instead of hydrolysis or substitution for the type of reaction.

**(d) (i)** Some very loosely drawn diagrams were unable to be given full credit: curly arrows should come from a clear origin (the middle of a bond, or from pair of electrons, rather than a negative charge) and, in cases such as this, the arrowhead needs to have both 'blades' and point to a specific atom or the middle of a bond.

Many candidates also confused partial charges/dipoles with full formal charges.

**(ii)** Candidates were required to reference heating the ethanolic NaOH to gain full credit.

**(iii)** Many answers retained the C=C or arranged the 3C monomer linearly with no branching. These were both common and recurring errors in questions of this type.

| | | |
|---|---|---|
| 3(a) | strong triple bond / high triple bond enthalpy | **1** |
| | non-polar (molecule) | **1** |

| | | |
|---|---|---|
| 3(b)(i) | *e.g.* $N_2 + 1½O_2 \rightarrow NO + NO_2$ | **1** |

**Question 3**

This question tested candidates' abilities to apply their knowledge in slightly unfamiliar contexts. In-depth knowledge and recall of nitrogen chemistry was crucial, as was the ability to draw clear, labelled, unambiguous diagrams.

**(a)** Different types of answer were needed to gain full credit. A strong triple bond and high $E_a$ are in effect making the same point.

**(b) (i)** There were many correct answers here, with balancing being the main difficulty.

**(ii)** Very few answers correctly referenced both acidic species, $HNO_3$ and $HNO_2$. The latter was rarely seen.

**(iii)** Candidates deployed a full range of answers referring to environmental issues: specifically, this was looking at photochemical smog.

**(c) (i)** Candidates found it challenging to apply the definition of standard enthalpy change of formation; attempting to conceive of two numbers that would sum to –46 (or –92), or that combined with the $\Delta H_f$ would sum to zero.

**(ii)** Many candidates were able to give good definitions here.

**(iii)** Candidates did not perform well on this question. Despite '**rate**' being in bold in the question, the large majority of responses attempted to answer in terms of yield and equilibrium position. Nevertheless, some credit was available for correct and non-contradicted information about a decrease in rate. Where collision theory is referenced, candidates are reminded of the need to discuss the *frequency* (i.e., number *per unit time*) of collisions, rather than simply the number.

**(d) (i)** Many diagrams showed a triple bond between the nitrogen atoms or miscounted the number of non-bonding electrons.

**(ii)** Candidates did not show a good understanding of hybridisation as responses divided reasonably equally between those who offered sp, $sp^2$ and $sp^3$ as answers.

**(iii)** Some good answers were seen here, but many answers could not be awarded full credit. *Sideways* overlap of p orbitals was needed in the response, or a clear and labelled diagram showing this as part of an unambiguous representation of a pi bond.

| | | |
|---|---|---|
| 4(a) | <table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table> | **1** |
| | <table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table> $CH_3CHBrCH_2CH_3$ | **1** |
| 4(b)(iii) | $C_4H_9CN + 2H_2O + H^+ \rightarrow C_4H_9COOH + NH_4^+$ | **1** |

**Question 4**

This an organic and spectroscopic question. Candidates are advised to be clear in matters of structure, both when representing molecules diagrammatically and when referring to features within a molecule, such as those identified by different types of spectra.

**(a)** This was answered well by many candidates. There were incidences of ambiguous answers, such as a 'crossed tick' (✓✗). Credit cannot be awarded if an answer is ambiguous.

**(b) (i)** Unambiguous structural formulae only were accepted, therefore molecular formulae were discounted and those in which the connectivity was questionable.

**(ii)** There were many correct answers to this question.

**(iii)** This proved to be one of the most challenging questions on the paper, requiring the balanced inclusion of (2)$H_2O$ and $H^+$ on the left-hand side, and $NH_4^+$ on the right. Salts from viable reactions of specific acids were accepted.

**(c) (i)** Candidates who understood the $M^+ : [M+1]^+$ ratio were generally able to gain full credit here. Occasional arithmetical errors were seen.

**(ii)** Many candidates gave 'ester' as a correct answer but were then unable to corroborate this with sufficient and necessary information from the spectrum. Reference to an absorption outside of the C=O ester region contradicted an answer; equally, an absorption within the 1710–1750 $cm^{-1}$ window was not solely sufficient, because of overlap with other absorptions. Corroboration could be given with reference to the C–O stretch (and the lack of O–H stretch).

C–H stretches are always ignored in these situations.

Absorptions must be accompanied by with assignment to an identified bond.

**(iii)** Propan-1-ol was a common incorrect answer for **T**. OH– is discouraged as a representation of an alcohol group. Many candidates did not draw the branched C5 acid correctly: some structures were unbranched; others had too many or too few H atoms.

| | | |
|---|---|---|
| 1(a)(i) | total # 3p $e^-$ 3 4 5 | **1** |
| | total # unpaired $e^-$ 3 2 1 | **1** |

| | | |
|---|---|---|
| 1(d)(i) | (relatively low melting and boiling points suggests) weak intermolecular / VdW forces | **1** |
| | (vigorous reaction with water suggests) **hydrolysis** | **1** |

| | | |
|---|---|---|
| 1(d)(ii) | $POCl_3 + 3H_2O \rightarrow H_3PO_4 + 3HCl$ | **1** |

**Question 1**

**(a) (i)** Most candidates were able to complete the table to show the number of electrons in the 3p subshells of P, S and Cl, as 3, 4 and 5 respectively.

The total number of unpaired electrons in these elements proved challenging for some candidates, Common incorrect answers were 1, 0 and 1 respectively.

**(ii)** The equation to represent the first ionisation of P was well answered. A common error was to omit the state symbols of (g) throughout.

**(ii)** The question required candidates to choose one of three possible values, given in the question, to represent the first ionisation energy of S and to explain their choice.

Most candidates did not mention that S has a pair of electrons in a 3p **orbital**, which experience spin-pair repulsion. This results in S having an ionisation energy value of 1000 kJ $mol^{-1}$, which is lower than both P and Cl. The explanation required candidates to recall that whilst S has a nuclear charge higher than P and lower than Cl, its nuclear attraction to the outer electrons is the lowest of these three elements.

Many candidates chose a value between P and Cl and supported their choice with a incorrect statement that the 'nuclear attraction to the outer electrons' increased across the period.

**(b) (i)** The full electronic configuration of the $P^{3-}$ ion was well answered as: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$.

**(ii)** The decrease in ionic radius from $P^{3-}$ to $S^{2-}$ and $Cl^-$ was well answered. Many correct explanations included mention of an increased attraction of the outer electron to the nucleus resulting from an increased nuclear charge with each ionic species having the same shielding.

**(c)** Observations were required for three tests carried out on separate samples of NaCl(aq).

When $Br_2$(aq) is added to a sample no reaction occurs, and the colourless solution of NaCl(aq) turns orange (because of the added $Br_2$).

Most answers stated that the observation from the addition of concentrated $H_2SO_4$ to NaCl(aq) is misty/steamy fumes, not recognising that the NaCl is in aqueous solution (and not a solid). The solution remains colourless as there is no visible reaction.

A white precipitate is formed when $AgNO_3$(aq) is added to NaCl(aq).

**(d) (i)** This question provided information that the chemical behaviour of $POCl_3$ is similar to that of $PCl_5$; $POCl_3$ has a low melting and boiling points and reacts vigorously with water suggesting that the structure and bonding of $POCl_3$ is simple covalent.

A small number of candidates gave a good explanation for this information and appreciated that the '*simple*' structure of $POCl_3$ was related to weak intermolecular forces of attractions and that the '*covalent*' nature of the compound is linked to the *hydrolysis* reaction with water.

**(ii)** The equation for the reaction of $POCl_3$ with water was generally quite well answered. Errors included an incorrect formula for $H_3PO_4$ and/or incorrect ratio between the water molecules and HCl product.

**(iii)** Most candidates were awarded full marks for the dot-and-cross diagram. A common error was the use of a dative bond between the P and O atoms.

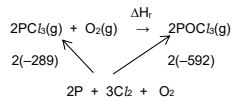
**(e) (i)** Most candidates were awarded the first mark for the definition of the enthalpy change of formation, for stating 'enthalpy change when 1 mole of a compound is formed'. However, very few correctly stated the second requirement that it is 'from its elements in their standard states'.

Many answers mentioned standard conditions which was not a requirement.

**(ii)** The calculation of the bond energy of P=O (in $PCl_3$), using $\Delta H_f$ and bond energy ($O_2$) data, proved challenging for most candidates.

Use of incorrect energy cycles was the major problem and full marks were rarely awarded.

A correct energy cycle is:




**$\Delta H_r$ = 2(–592) – 2(–289) = -606 kJmol$^{-1}$**

| bonds broken | bonds formed |
|---|---|
| ~~6 P-Cl~~ | ~~6 P-Cl~~ |
| 1 O=O (496) | 2 P=O |

496 - 2P=O = -606
-2P=O = -1102
**P=O = 551kJmol$^{-1}$**

| | | |
|---|---|---|
| 2(a)(i) | increases down the group | **1** |
| 2(a)(iv) | $Ba(CH_3COO)_2$ / barium ethanoate | **1** |
| 2(a)(v) | $Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$ | **1** |
| 2(b)(i) | $CaCO_3$ **AND** thermal stability of carbonates **increases** down the group | **1** |
| 2(b)(ii) | $BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$ | **1** |

**Question 2**

**(a) (i)** The solubilities of group 2 hydroxides 'increases' down the group; this variation in solubilities was correctly stated by many candidates

**(ii)** When barium oxide is added to water the observation is that the white solid dissolves/disappears to form a colourless solution. A common error was that there was bubbling/effervescence seen.

**(iii)** In the given reaction scheme, reaction 2 involved the formation of $BaCO_3$ and water from adding a reactant to $Ba(OH)_2$. Carbon dioxide, $CO_2$, is a reactant that would form both the carbonate and water. Many answers included a metal carbonate which would not give water as a product.

**(iv)** The addition of ethanoic acid to $Ba(OH)_2$ gave barium ethanoate, $(CH_3COO)_2Ba$, as the product.

Many candidates appreciated that barium ethanoate was formed but opted to answer using a formula of $(CH_3COO)Ba$, which was not creditworthy.

**(v)** $Ba(OH)_2$ and hydrogen are formed when Ba is added to water and the equation is:

$$Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$$

A common error was to use only 1 mole of water molecules and water formed as a product.

**(b) (i)** The thermal stability of group 2 carbonates increases down the group; hence, when barytocalcite, $BaCa(CO_3)_2$, is heated $CaCO_3$ will decompose first. Most answers correctly stated that $CaCO_3$ would decompose first but explanations compared the reactivity of Ca with Ba.

**(ii)** The equation to represent the thermal decomposition of $BaCa(CO_3)_2$ was often correct.

$$BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$$

Common errors included answers with BaCa as a product or omitting $2CO_2$ as a product.

**(c)** Full credit for the hydrolysis reaction was rarely awarded. The two organic products are methanol, $CH_3OH$, and the diacid.

That both the –CN and $–CO_2CH_3$ groups will undergo hydrolysis was not recognised by many candidates. The –CN group was frequently left unhydrolysed and the product methanol was not given as the second organic product.

| | | |
|---|---|---|
| 3(a) | compound / molecule / substance that oxidises another / is reduced | **1** |

| | | |
|---|---|---|
| 3(d)(i) | $pV = nRT \therefore V = 8.31 \times 630 \div 100\,000 =$ **<u>0.0524</u>** ($m^3$) | **1** |

| | | |
|---|---|---|
| 3(d)(ii) | moles of gas = $12 \times 5.00 / 180$ | **1** |
| | volume of gas = M1 × **(d)(i)** = **0.0175** ($m^3$) | **1** |

**Question 3**

**(a)** The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

**(b)** This question required the completion of an energy diagram/energy profile to represent the catalytic effect of $MnO_2$(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

**(c)** The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and $KCl$, in which the $Cl$ is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

**(d) (i)** PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 $m^3$, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

**(ii)** This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous $CO_2$ and $H_2O$ in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of $CO_2$ and $H_2O$ produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

**(e) (i)** A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

**(ii)** This question required observations for three tests.

The observation for a reaction of glucose + warm, acidified $KMnO_4$(aq) is that the solution changes from purple to colourless.

Many candidates were given credit for a correct observation.

Glucose + warm, Fehling's reagent gave an orange/red precipitate which again was recognised by many candidates. Several candidates omitted the precipitate or stated that a silver mirror formed.

Glucose + warm, alkaline $I_2$(aq) was incorrectly stated as a yellow precipitate by many candidates. There is no reaction with glucose, so the solution remains colourless.

**(iii)** Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

| | | |
|---|---|---|
| 2(a) | curved line from (0,0) to reach same horizontal line | **1** |
| 2(e)(i) | species with 1 or more unpaired electrons | **1** |
| 2(e)(ii) | $NO_2 + SO_2 \rightarrow SO_3 + NO$ | **1** |
| | $NO + ½ O_2 \rightarrow NO_2$ | **1** |
| 2(f)(i) | magnesium hydroxide / $Mg(OH)_2$ | **1** |
| 2(f)(ii) | $Mg(NO_3)_2 \rightarrow MgO + 2NO_2 + ½O_2$ | **1** |

**Question 2**

**(a)** Some candidates appeared to confuse graphs of rate versus time with those showing concentration versus time.

**(b)** Well-rehearsed arguments related the significance of increasing temperature on the position of equilibrium to the example given. More able candidates showed no ambiguity and made explicit reference to the effect of an increase in temperature on the position of this equilibrium in terms of either the '*forward reaction is exothermic'* or '*the reverse reaction is endothermic'*. Less precise answers included '*the reaction moves backwards as it's an exothermic reaction'*.

**(c)** This question proved to be demanding. A few candidates found $K_p$ using an appropriate method; they worked out the mole fraction of the components, converted them into partial pressures and then substituted these values into the correct expression for $K_p$. Weaker candidates calculated $K_p$ directly from moles and included moles in the units. Deduction of units also presented difficulties. Many answers gave units in terms of Pa with no attempt to convert kPa to Pa within the calculation.

**(d)** Lightning was well known as a natural process responsible for formation of atmospheric NO and $NO_2$. 'Catalytic converter' as the man-made process was a common incorrect answer.

**(e) (i)** Many correct definitions of the term free radical were seen. Ambiguous descriptions referred to '*unpaired electron'* as '*unshared*' or '*unbonded*' or '*free*'. Common incorrect answers referred to a free radical as *'a species with a lone pair of electrons'*.

**(ii)** The pair of equations which showed the catalytic role of $NO_2$ in the oxidation of $SO_2$ were not well known. The equation to represent the regeneration of $NO_2$ appeared to cause most difficulties with many incorrectly balanced equations seen.

**(iii)** The environmental consequence of the oxidation of sulfur dioxide was well known.

**(f) (i)** Identification of a magnesium containing base was relatively common. Incorrect answers included salts, rather than bases, which contained magnesium or bases that were not made from magnesium. The formula of the compound identified was not always correct.

**(ii)** This equation proved challenging for many. Some answers incorrectly balanced equations which included all correct species. Use of an incorrect formula for magnesium nitrate was a common error, including $Mg_3N_2$.

**(iii)** Identification of the change in thermal stability down Group 2 nitrates was fairly well known.

| | | |
|---|---|---|
| 3(a)(i) | $4P / P_4 + 5O_2 \rightarrow P_4O_{10}$ | **1** |
| | $P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$ | **1** |

| | | |
|---|---|---|
| 3(d)(i) | $H_3PO_4 + 3CH_3OH \rightarrow (CH_3O)_3PO + 3H_2O$ | **1** |

**Question 3**

**(a) (i)** Fully balanced equations were seen by more able candidates.

**(ii)** This definition was described accurately in most candidates answers. A common misconception was seen in weaker responses that described a weak acid as one that *'did not completely dissolve'*.

**(b) (i)** Many candidates gave a definition of the enthalpy change which occurs when 'one mole of product is formed from its elements'. It was less common for the elements to be described as 'in their standard states'.

**(ii)** More able candidates gave excellent and logical workings that contained clearly labelled energy cycles. It was common to see an expression for the different bond energies added or subtracted from each other, in a variety of possible incorrect combinations, with no attempt to show how these expressions were deduced. The stoichiometry of the equation was ignored by some.

**(iii)** Excellent answers were seen which identified phosphorus in $H_3PO_3$ as both oxidised and reduced and linked the process of oxidation and reduction to the relevant changes in oxidation number for the three phosphorus containing species. Weaker responses were less specific, for example '*$H_3PO_3$ is oxidised and reduced'*. Oxidation numbers were sometimes described incorrectly.

**(c) (i)** Some candidates identified **A** as steam or $H_2O(g)$, other candidates were less precise and identified **A** as water. A common incorrect answer was NaOH.

**(ii)** Many unambiguous representations of 2-bromopropane were seen. Occasionally the central carbon atom was drawn with a valency of 3.

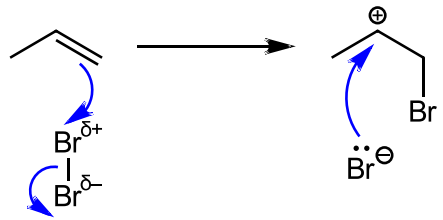
**(iii)** Description of this esterification reaction as a condensation reaction was relatively well known. Incorrect answers included hydrolysis and oxidation.

**(iv)** This question proved to be demanding. Candidates were expected to ignore all bonds which were common to both reactant and product. While the appropriate bond, C=O, was identified in many answers C-O and C-H were also seen. Most candidates only identified a range of wavenumbers where a change in absorption occurs rather than described if the absorption appeared, disappeared or moved.

**(d) (i)** More able candidates deduced a formula for the correct organic product in the equation. The inclusion of $H_2O$ as the additional product was less common, even when included the equation was not always balanced.

**(ii)** The appropriate formula to deduce the number of carbon atoms present in **T** was not well known and many confused calculations were seen. It was common for candidates to give a structural formula for **T** rather than the required molecular formula.

| | | |
|---|---|---|
| 4(a)(i) | (it has molecules that) contain a chiral carbon / centre **OR** are non-superimposable mirror images | **1** |

| | | |
|---|---|---|
| 4(b)(i) |  <br>curly arrow from C=C bond to $Br^{\delta+}$ **AND** curly arrow from $Br^{\delta+}$— $Br^{\delta-}$ bond to $Br^{\delta-}$ **AND** $Br^{\delta+}$ closest to the double bond | **1** |
| | correct intermediate | **1** |
| | curly arrow from lone pair on $Br^-$ to $C^{(+)}$ of the intermediate | **1** |

| | | |
|---|---|---|
| 4(b)(ii) | $CH_3CH(OH)COOH + [O] \rightarrow CH_3COCOOH + H_2O$ | **1** |

**Question 4**

**(a) (i)** Some candidates gave clear explanations of the presence of a chiral carbon in the lactic acid structure. A few candidates stated that optical isomers had chiral centres or described the behaviour of an optical isomer in the presence of polarised light.

**(ii)** Many candidates were unable to give the systematic name of lactic acid. 'Carboxylic acid' was seen frequently.

**(iii)** It was common for diagrams to show **more** than one water molecule or **multiple** hydrogen bonds. Candidates should be advised to respond as directed by the question. It was quite common for answers, showing more than one hydrogen bond, to contradict a correct representation.

**(b) (i)** Some clear and precise reaction mechanisms were shown with partial charges assigned to relevant atoms and unambiguous representation of curly arrows to show bonds broken and bonds made. When drawing mechanisms of this type candidates should be encouraged to consider what the curly arrows represent and whether their representation is clear, use partial charges correctly and discriminately and check that the number of carbon atoms on the intermediate is correct.

**(ii)** The equations described sometimes contained only pyruvic acid as a product or identified $H_2$ rather than $H_2O$ as a product.

**(iii)** This question was demanding. Many candidates confused the reagents and conditions required to convert halogenoalkanes to nitriles with those required to convert ethanal to a hydroxynitrile. Deduction of the reagents and conditions used in the second step of both synthesis from propene and ethanal proved demanding for most candidates. Oxidising agents were more commonly seen for the third steps, however, some candidates represented the oxidising agents with incorrect formulae or with no reference to acid conditions.

| | | |
|---|---|---|
| 1(a)(i) | total # 3p $e^-$ 3 4 5 | **1** |
| | total # unpaired $e^-$ 3 2 1 | **1** |

| | | |
|---|---|---|
| 1(d)(i) | (relatively low melting and boiling points suggests) weak intermolecular / VdW forces | **1** |
| | (vigorous reaction with water suggests) **hydrolysis** | **1** |

| | | |
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**Question 1**

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Most candidates did not mention that S has a pair of electrons in a 3p **orbital**, which experience spin-pair repulsion. This results in S having an ionisation energy value of 1000 kJ $mol^{-1}$, which is lower than both P and Cl. The explanation required candidates to recall that whilst S has a nuclear charge higher than P and lower than Cl, its nuclear attraction to the outer electrons is the lowest of these three elements.

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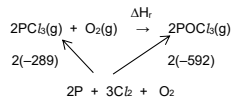
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|---|---|
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| 1 O=O (496) | 2 P=O |

496 - 2P=O = -606
-2P=O = -1102
**P=O = 551kJmol$^{-1}$**

| | | |
|---|---|---|
| 2(a)(i) | increases down the group | **1** |
| 2(a)(iv) | $Ba(CH_3COO)_2$ / barium ethanoate | **1** |
| 2(a)(v) | $Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$ | **1** |
| 2(b)(i) | $CaCO_3$ **AND** thermal stability of carbonates **increases** down the group | **1** |
| 2(b)(ii) | $BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$ | **1** |

**Question 2**

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Many candidates appreciated that barium ethanoate was formed but opted to answer using a formula of $(CH_3COO)Ba$, which was not creditworthy.

**(v)** $Ba(OH)_2$ and hydrogen are formed when Ba is added to water and the equation is:

$$Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$$

A common error was to use only 1 mole of water molecules and water formed as a product.

**(b) (i)** The thermal stability of group 2 carbonates increases down the group; hence, when barytocalcite, $BaCa(CO_3)_2$, is heated $CaCO_3$ will decompose first. Most answers correctly stated that $CaCO_3$ would decompose first but explanations compared the reactivity of Ca with Ba.

**(ii)** The equation to represent the thermal decomposition of $BaCa(CO_3)_2$ was often correct.

$$BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$$

Common errors included answers with BaCa as a product or omitting $2CO_2$ as a product.

**(c)** Full credit for the hydrolysis reaction was rarely awarded. The two organic products are methanol, $CH_3OH$, and the diacid.

That both the –CN and $–CO_2CH_3$ groups will undergo hydrolysis was not recognised by many candidates. The –CN group was frequently left unhydrolysed and the product methanol was not given as the second organic product.

| | | |
|---|---|---|
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| | | |
|---|---|---|
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| | | |
|---|---|---|
| 3(d)(ii) | moles of gas = 12 × 5.00 / 180 | **1** |
| | volume of gas = M1 × **(d)(i)** = **0.0175** ($m^3$) | **1** |

**Question 3**

**(a)** The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

**(b)** This question required the completion of an energy diagram/energy profile to represent the catalytic effect of $MnO_2$(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

**(c)** The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and KCl, in which the Cl is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

**(d) (i)** PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 $m^3$, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

**(ii)** This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous $CO_2$ and $H_2O$ in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of $CO_2$ and $H_2O$ produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

**(e) (i)** A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

**(ii)** This question required observations for three tests.

The observation for a reaction of glucose + warm, acidified $KMnO_4$(aq) is that the solution changes from purple to colourless.

Many candidates were given credit for a correct observation.

Glucose + warm, Fehling's reagent gave an orange/red precipitate which again was recognised by many candidates. Several candidates omitted the precipitate or stated that a silver mirror formed.

Glucose + warm, alkaline $I_2$(aq) was incorrectly stated as a yellow precipitate by many candidates. There is no reaction with glucose, so the solution remains colourless.

**(iii)** Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

| | | |
|---|---|---|
| 1(a)(i) | giant metallic | **1** |
| 1(b)(i) | (the mass of) one twelfth of the mass of a C-12 / $^{12}C$ atom (or isotope) | **1** |
| 1(b)(ii) | average mass of the isotopes of an element compared to the unified atomic mass unit | **1** |
| 1(d)(i) | $2Cu^{2+} + 4I^{-} \rightarrow 2CuI + I_2$ | **1** |

**Question 1**

**(a) (i)** A high proportion of candidates knew that copper is metallic. Many candidates omitted that the lattice is giant.

**(ii)** Diagrams showing the metallic bonding as an array of positive metal ions surrounded by delocalised electrons were varied, with many either omitting or not correctly labelling the positive ions and delocalised electrons.

**(b) (i)** Some candidates incorrectly referred to a mole of $^{12}C$ atoms or just an atom of $^{12}C$.

**(ii)** Many correct answers were seen. Some candidates omitted the word 'average'.

**(iii)** The question was quite well answered. The major error included the use of 63 and 65 as the masses of the copper isotopes instead of the data given in the question.

**(c)** Very few candidates recognised that the ion with an abundance of 23% and an *m/e* of 31.5 was that of $^{63}Cu^{2+}$.

**(d) (i)** The equation was not well answered overall. Few candidates recognised that it was a redox reaction. Many answers contained electrons on one side of the equation and/or were completely unbalanced.

**(ii)** The identification of $Cu^{2+}$ as the oxidising reagent in this reaction was generally not well known. Many candidates answered with just 'copper'.

**(iii)** The full electronic configuration of $Cu^{2+}$ was frequently incorrect with the $4s^2$ electrons still present in many answers.

| | | |
|---|---|---|
| 2(a)(i) | it / aluminium oxide is insoluble (in water)<br>**OR**<br>it / aluminium oxide does not react (with water) | **1** |
| 2(a)(ii) | **M1** correct state of correct aluminium product<br>**M2** balanced equation for reaction of aluminium oxide with NaOH<br>$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4(aq)$<br>**OR**<br>$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2(aq) + H_2O$ | **2** |
| 2(b) | $Mg(NO_3)_2$ | **1** |
| 2(c)(i) | Al (+)3 / III **AND** P (+)5 / V | **1** |
| 2(c)(iii) | $PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$ | **1** |

**Question 2**

**(a) (i)** Many candidates knew that aluminium objects covered in a layer of aluminium oxide do not react with cold water because the aluminium oxide is insoluble in water and does not react with the water.

**(ii)** The equation, with state symbols, was quite well answered, with many candidates recognising that the product was aqueous sodium aluminate.

**(iii)** Some good answers were seen.

**(b)** Some candidates gave the formulae of nitrates of elements that are not located in Period 3.

**(c) (i)** Several candidates correctly stated the oxidation number of Al in $AlCl_3$. The answers for that of P in $PCl_5$ were varied.

**(ii)** Many candidates knew that Al and P had different numbers of electrons available to use. Many omitted where these electrons are located.

**(iii)** It was well recognised that $PCl_5$ was the required chloride of phosphorus and that HCl is a product of this reaction. Few candidates gave the correct product containing phosphorus. Many answers contained an oxide of phosphorus and/or an incorrect phosphoric acid.

**(iv)** Some candidates knew the reaction is a hydrolysis reaction with other candidates correctly describing it as a substitution reaction.

**(v)** Some good answers were seen.

| | | |
|---|---|---|
| 3(a) | **M1** (energy change) when 1 mole of water is formed<br>**M2** from an (aqueous) acid and an alkali / aqueous base | **2** |
| 3(b)(i) | 0.05 mol ($H_2SO_4$(aq)) | **1** |
| 3(c)(i) | $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$ | **1** |

**Question 3**

**(a)** Many candidates incorrectly stated, 'enthalpy change when 1 mole of acid reacted with 1 mole of base'.

**(b) (i)** Many candidates calculated the initial amount of $H_2SO_4$ instead of calculating the amount of acid that was neutralised by the NaOH.

**(ii)** Very few candidates deduced that Q = 5710 J and the mass of the solution was often incorrectly given as either 1.00 g or a number derived from calculating the masses of NaOH and $H_2SO_4$ in the reaction.

**(c) (i)** Some good answers were seen.

**(ii)** Several candidates knew that $CH_3COOH$ was not fully dissociated, as it is a weak acid, compared to the strong acid HC*l*, which resulted in a less negative value for the $\Delta H_{neut.}$ Very few candidates knew that the difference in values of the respective $\Delta H_{neut}$ values was due to the additional energy required to fully dissociate the weak acid such that less energy is released after neutralisation.

| | | |
|---|---|---|
| 4(a)(i) | $2KCl + H_2SO_4 \rightarrow K_2SO_4 + 2HCl$<br>**OR**<br>$KCl + H_2SO_4 \rightarrow KHSO_4 + HCl$ | **1** |

**Question 4**

**(a) (i)** Some good answers were seen.

**(ii)** Very few candidates knew that hydrogen iodide is not prepared in the same way as hydrogen chloride because the hydrogen iodide and concentrated $H_2SO_4$ would undergo a redox reaction to form a variety of products to include $SO_2$, $H_2S$ and $I_2$.

**(b) (i)** A small number of candidates recognised that when a sample of colourless HI(g) is placed in a sealed container and allowed to reach equilibrium the colour changes to purple, attributed to the formation of $I_2$(g).

**(ii)** This question was well answered by many candidates.

**(iii)** The calculation of the concentration of iodine in an equilibrium mixture was generally not well answered. Candidates who deduced that [HI] is (1.70 ÷ 2 =) 0.85 mol in the equilibrium mixture tended to gain some credit. Very few used the expression $\sqrt{(0.0217 \times 0.85)}$ to calculate the $[I_2]$ as 0.125 mol $dm^{-3}$.

**(c) (i)** Some candidates were able to describe the difference in composition of the equilibrium mixture at the two temperatures.

**(ii)** Some candidates deduced that the reaction is endothermic in the forward direction leading to a lower concentration of $H_2$ and $I_2$ in the equilibrium mixture.

| | | |
|---|---|---|
| 5(a) | $C_3H_7O$ | **1** |

| | | |
|---|---|---|
| 5(b)(i) | $C_6H_{10}O_3 + [6H] \rightarrow C_6H_{14}O_2 + H_2O$ | **1** |

**Question 5**

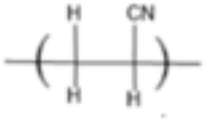
**(a)** Some candidates deduced the correct empirical formula of $C_3H_7O$ from a molecular formula of $C_6H_{14}O_2$.

**(b) (i)** Very few candidates used molecular formula, as required in the question, to construct an equation for the reduction reaction.

**(ii)** Many answers correctly stated that a suitable reducing agent for the reaction is $LiAlH_4$.

**(c)** Some good answers were seen.

**(d)** Many candidates correctly stated that **X**, the ketoacid, would react with all three reagents to give effervescence, an orange precipitate and yellow precipitate respectively. Fewer candidates recognised that the diol, **Y**, would only react with the alkaline $I_2(aq)$ as it contained a $CH_3CH(OH)$ moiety.

| | | |
|---|---|---|
| 6(a) |  OR  | **1** |

| | | |
|---|---|---|
| 6(e) | $CH_3CH_2CN + 2H_2 \rightarrow CH_3CH_2CH_2NH_2$ | **1** |

**Question 6**

**(a)** The single repeat unit derived from the monomer, $CH_2$=CHCN, by addition polymerisation was quite well answered.

**(b) (i)** Many candidates recognised that the mechanism for step 1 was nucleophilic addition.

**(ii)** Overall, the mechanism for step 1 was generally not answered well. The polarity on the C=O bond was often absent and the arrow from the $CN^-$ to the C (in the C=O bond) did not start from the lone pair on the C of the $CN^-$ ion. Several answers contained an incorrect intermediate with a positive charge which was then attacked by the $CN^-$ nucleophile and many answers incorrectly showed an arrow from an $H^+$ ion to an $O^-$ (in the intermediate).

**(iii)** Some good answers were seen.

**(iv)** The absorptions in the infrared spectrum of $CH_2$=CHCN were often correctly identified as C≡N and C=C. Incorrect answers mainly contained an identification of the functional groups rather than the bond or identifying the absorption (annotated as T on the IR spectrum) as that of a C=O group.

**(c) (i)** Very few candidates gave an accurate description.

**(ii)** A number of answers were awarded some credit. Few candidates were able to give a full description.

**(d)** Several candidates gave a correct pair of molecules that were non-superimposable images of $CH_3CH(OH)CN$. A common error was diagrams containing the same type of bond at 180° to each other.

**(e)** The equation for the reduction of propanenitrile to propylamine using hydrogen gas was not well known. Few candidates used the correct stoichiometry for hydrogen.

**(f) (i)** It was well known that the name of the type of reaction in step 1 is substitution.

**(ii)** Many candidates gained credit for ammonia. Many omitted heat or the solvent as conditions.

| | | |
|---|---|---|
| 2(a)(i) | <table><tr><td>element</td><td>Na</td><td>Mg</td><td>Al</td><td>Si</td><td>P</td></tr><tr><td>maximum oxidation number of elements in chlorides</td><td>(+)1</td><td>(+)2</td><td>(+)3</td><td>(+)4</td><td>(+)5</td></tr></table>All correct for one mark | **1** |
| 2(b)(i) | $SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$ | **1** |
| 2(c)(i) | $PCl_5 + 4H_2O \rightarrow 5HCl + H_3PO_4$ | **1** |
| 2(d)(i) | $Cl_2 + H_2O \rightleftharpoons HOCl + HCl$ | **1** |
| 2(d)(ii) | $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$ | **1** |
| 2(e)(ii) | $NaClO + H_2O_2 \rightarrow O_2 + NaCl + H_2O$ | **1** |
| 2(e)(iii) | **M1** *(show how to calculate) mol $O_2$ gas*<br>42 $cm^3$ gas = 42 / 24000 mol oxygen gas (= $1.75 \times 10^{-3}$ mol)<br>**M2** *(process to find) amount NaClO in 10 $cm^3$ sample = 100 $cm^3$ (diluted) bleach*<br>M1 x 4 (= $4 \times 1.75 \times 10^{-3}$ mol in 10 $cm^{-3}$ = 0.007)<br>**M3** correct calculation using M2 to find concentration<br>= (M2 /10) $\times$ 1000 $\times$ 74.5 = 52.2 ($g / dm^3$) | **3** |

**Question 2**

**(a) (i)** The maximum oxidation numbers of the Period 3 chlorides were well known.

**(ii)** The majority of responses were correct. Some candidates produced responses that were vague and incomplete. Occasionally, explanations were incorrectly given in terms of changes to the nuclear charge.

**(b) (i)** The majority of candidates identified the correct formula of the silicon chloride and gave an appropriate balanced equation for the reaction.

**(ii)** A significant proportion of candidates incorrectly described the pH of the solution as greater than or equal to 7.

**(c) (i)** The equation to represent the reaction for phosphorus(V) chloride with water was less well known than the equation for **(b)(ii)**. The production of phosphorus oxide rather than phosphoric acid was a common incorrect answer. Other equations which showed appropriate species were not correctly balanced.

**(ii)** Many candidates gave appropriate suggestions for the pH of the solution produced in **(c)(i)**.

**(d) (i)** Equations using C*l* rather than C*l*$_2$ and production of $O_2$ with HC*l* were common incorrect answers.

**(ii)** A small proportion of candidates identified all the correct species and produced a balanced equation. Some candidates found the equation for the reaction of chlorine with hot NaOH(aq) to be demanding.

**(e) (i)** A wide range of incorrect species were seen in the answers; many of which were not ions.

**(ii)** Many correct equations representing this reaction were seen. Occasionally, equations gave incorrect multiples for the correct species which included the decomposition of some hydrogen peroxide without the use of sodium chlorate(I).

**(iii)** A significant proportion of candidates used the appropriate data to calculate the number of moles of sodium chlorate(I) used in a single titration. Difficulties arose in attempts to work out the concentration in the original sample, **W**. Occasionally, concentrations were left in mol dm$^{-3}$ rather than converting this into g dm$^{-3}$, as detailed in the question.

| | | |
|---|---|---|
| 3(a)(i) | KCN in ethanol + heat | **1** |

| | | |
|---|---|---|
| 3(a)(iv) | 2 $CH_3CH_2COOH + Ca \rightarrow (CH_3CH_2COO)_2Ca + H_2$<br>**M1** $H_2$ as product<br>**M2** balanced equation based on correct formula of Y | **2** |

**Question 3**

**(a) (i)** Some candidates incorrectly described the reagent and conditions required to convert carbonyl compounds to hydroxynitriles, rather than halogenoalkanes to nitriles.

**(ii)** The systematic name for **X** was not always used, e.g. propionitrile or ethyl cyanide. Omission of the 'e' in propanenitrile was often seen.

**(iii)** It was generally well known that this type of reaction is a hydrolysis reaction.

**(iv)** Many candidates correctly produced equations representing the production of hydrogen gas. The formula of the calcium salt was frequently given in terms of calcium with a valency of one.

**(b)** Many candidates knew that acidic conditions were required. Stronger candidates appreciated that the essential condition for this reaction was 'heat under reflux' to ensure complete oxidation to the carboxylic acid.

**(c)** Most candidates described the correct number of pi bonds present in **X**; the correct number of sigma bonds was seen less frequently. Seven, nine and ten sigma bonds were common incorrect answers.

| | | |
|---|---|---|
| 1(a) | <table><tr><td></td><td>nucleon number</td><td>number of neutrons</td><td>number of electrons</td></tr><tr><td>Tellurium-130</td><td>130</td><td>78</td><td>52</td></tr></table> | **3** |
| 1(e) | **M1** 6 bonding pairs (of electrons and 0 lone pairs) **OR** coordination number = 6 **OR** $x = 6$<br>**M2** bonding pairs repel (equally) | **2** |
| 1(f)(i) | $TeF_x + xH_2O \rightarrow Te(OH)_x + xHF$ **OR** $TeF_6 + 6H_2O \rightarrow Te(OH)_6 + 6HF$ | **1** |

**Question 1**

**(a)** This was generally well answered. Some candidates were not familiar with the term nucleon number, often interpreting this as neutron number.

**(b)** Many candidates answered this question correctly. Common errors included not specifying **1**s and giving 5p as the answer.

**(c)** Candidates generally answered this well. State symbols were not always included, especially in $Te^{+}$.

**(d) (i)** The majority of candidates correctly identified two factors. Many candidates incorrectly stated that the shielding effect remains constant. An 'increase in nuclear charge' was a common incorrect response that negated other descriptions.

**(ii)** Most candidates recognised a general increase in successive ionisation energies, with a majority of those able to indicate the most significant jump from the $6^{th}$ to $7^{th}$ ionisation energy. Many answers incorrectly included a graph with a dip, most often in energy at the $6/7^{th}$ ionisation energy.

**(e)** Candidates who correctly referenced spin-pair interactions gained credit. Few candidates could use the VSEPR theory correctly.

**(f) (i)** This was generally not answered well. Many responses included a mixture of numbers and '*x*'s. A small proportion of candidates gave a correctly balanced equation.

**(ii)** Many candidates correctly identified this as a 'hydrolysis' reaction. A common answer was 'displacement', which was not enough to gain credit.

| | | |
|---|---|---|
| 2(a) | **M1** (enthalpy change) when 1 mole of water is formed<br>**M2** from an (aqueous) acid and an alkali / (aqueous) base | **2** |

| | | |
|---|---|---|
| 2(b)(i) | 0.05 mol | **1** |

| | | |
|---|---|---|
| 2(c)(i) | ..(1)..$H_2SO_4$(aq) + ..(1)..$Ba(OH)_2$(aq) → …(1)..$BaSO_4$(s) + $2H_2O$(l)……<br>**M1** correct state symbols for both products<br>**M2** correctly balanced equation | **2** |

**Question 2**

**(a)** Some candidates were able to give a concise and accurate definition. Overall, the recall of this definition was not well attempted. Candidates often omitted any reference to a molar scale, although 'one mole of acid reacting with one mole of base' was commonly seen as an inadequate response. A common misconception was that neutralisation requires energy.

**(b) (i)** Few candidates were able to incorporate the relevant stoichiometry involved into their calculations.

**(ii)** Very few candidates gained full credit here. A very small proportion of candidates correctly used 175 g as the correct mass, many instead quoting 1 g. Many candidates converted the temperature to Kelvin but erroneously adding 273 to 7.8. A very small proportion of candidates correctly divided the energy released by the number of moles of water formed.

**(c) (i)** This was well answered by most candidates. Most were able to provide a correct balanced equation. Some gave the incorrect state of barium sulfate.

**(ii)** Many candidates stated that $BaSO_4$ is insoluble, though not many were able to link this to an additional energy change. Several answers incorrectly referred to the solubility of $Ba(OH)_2$.

| | | |
|---|---|---|
| 3(a) | **M1** *appearance* (colourless / yellow) liquid<br>**M2** *structure and bonding* simple / molecular **AND** covalent | **2** |

| | | |
|---|---|---|
| 3(e) | $CHCl_3 + 2HF \rightarrow CHClF_2 + 2HCl$ | **1** |

**Question 3**

**(a)** Very few candidates could state the appearance of $SiCl_4$ at room temperature; more were able to state the correct structure and bonding. The most common incorrect response was giant covalent.

**(b)** Most candidates correctly referred to the appearance and pH. Candidates often did not refer to the relevant observation and struggled to differentiate the behaviours of the two chlorides.

**(c) (i)** A large number of candidates stated the correct reagent. Many left out aqueous or dilute as the conditions.

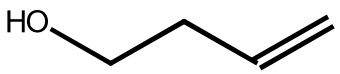
**(ii)** Many of the candidates defined disproportionation; a small proportion were able to refer to the correct oxidation numbers. Several referred incorrectly to the +1 oxidation number for chlorine.

**(d) (i)** This was well answered by most candidates.

**(ii)** Most candidates correctly answered the initiation step. Many candidates found the propagation and termination steps more challenging.

**(e)** This was well answered by most students.

**(f)** Many candidates were able to give the correct formulae at $m/e$ = 164.Very few candidates gave both of the correct options at $m/e$ = 166.

| | | |
|---|---|---|
| 4(a)(i) |  | **1** |
| 4(b)(i) | cold dilute acidified $KMnO_4$ | **1** |
| 4(b)(ii) | $CH_2O$ | **1** |

**Question 4**

**(a) (i)** Common errors were to retain the COOH group and to have the C=C double bond reduced.

**(ii)** This was well answered by most candidates.

**(b) (i)** This was generally well answered, although few candidates stated all the required conditions.

**(ii)** This was well answered by most candidates.

**(iii)** A small proportion of candidates gained full credit here.

**(c)** Many candidates correctly gave pentan-1-ol. A small number of candidates made the link between the yellow precipitate with alkaline $I_2$(aq) and a $CH_3CH(OH)$– group. Many candidates struggled to synthesise all the detail to present a cogent answer to the question.

| | | |
|---|---|---|
| 5(a)(i) | $C_8H_{18} + 12½\ O_2 \rightarrow 8CO_2 + 9H_2O$<br>**M1** correct formula of M $C_8H_{18}$<br>**M2** balanced equation showing complete combustion of M | **2** |

| | | |
|---|---|---|
| 5(b)(i) | **M1** ● high temperature ● high pressure ● $N_2$ Any two correct for **M1**<br>**M2** $N_2 + O_2 \rightarrow 2NO$ **OR** $N_2 + 2O_2 \rightarrow 2NO_2$ **OR** $N_2 + xO_2 \rightarrow 2NO_x$ | **2** |

**Question 5**

**(a) (i)** This was well answered by most candidates, although balancing the equation proved challenging for some.

**(ii)** Most candidates correctly mentioned the production of CO instead of $CO_2$. A very small number correctly referred to C and unburnt hydrocarbons.

**(b) (i)** A large proportion of candidates incorrectly interpreted this as a question about catalytic converters. Many answers used NO instead of $N_2$.

**(ii)** A small number of candidates correctly identified the catalytic role of $NO_x$ and gave all the relevant equations. Many gave the equation for the formation of $H_2SO_4$, and a significant number gave equations and answers related to the formation of $HNO_3$ for acid rain.

**(iii)** This was well answered by most. Some candidates gave the identity of a gas instead of the type of air pollution.

**(c) (i)** Candidates often omitted dilute/(aq) with a named acid here.

**(ii)** A few candidates recognised that all three ester groups will undergo hydrolysis to form propanetriol. Many answers gave fragments where only one ester group had undergone hydrolysis.

**(iii)** This was well answered by most candidates. The majority of answers referred to esterification/substitution instead of condensation.

**(iv)** A small proportion of candidates answered this question correctly.

**(v)** A small proportion of candidates answered this question correctly.

| | | |
|---|---|---|
| 1(a)(i) | power of an atom to attract electrons to itself | **1** |
| 1(d)(i) | $SO_2 + H_2O \rightarrow H_2SO_3$ | **1** |
| 1(d)(ii) | $SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$ | **1** |
| 1(d)(iii) | $CO_2 + Mg(OH)_2 \rightarrow MgCO_3 + H_2O$ | **1** |

**Question 1**

This question tested ideas of atomic structure, periodicity and some quantitative chemistry. Knowledge of electronegativity was less secure and many Period 3 equations were incorrectly presented as redox reactions.

**(a) (i)** There were some very good answers here, although many candidates tried to answer in terms of ionisation or electron affinity.

**(ii)** Few candidates were awarded both marks as this required a balance of points to construct an argument. Spin-pair repulsion arguments were unnecessarily presented on a number of occasions. Many mentioned the distance of the outermost electron from the nucleus without any reference to its significance. Other answers included non-justified references to ionic radius or ionisation energy.

**(b) (i)** A whole array of different diagrams was seen, with only the ionic form ($Li^+$ and $H^-$) acceptable. All electrons, only four in total, were required to be shown.

**(ii)** Linear, trigonal planar and planar were popular incorrect answers.

**(c) (i)** The most common error here was the omission of state symbols from the elemental species. Many answers suggested that $H_2$ was being ionised.

**(ii)** This was well answered by many candidates.

**(iii)** Most candidates answered this well, although $3p^6$ was commonly seen.

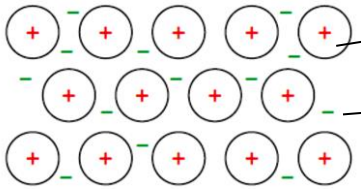
**(d) (i)** A large number of answers erroneously suggested $H_2SO_4$ as a product.

**(ii)** Both $Na_2S$ and $Na_2SO_4$ appeared regularly, as did $H_2$. Candidates are advised to consider whether redox has occurred in these reactions.

**(iii)** Many incorrect versions for the formula of the carbonate were seen such as $Mg(CO_3)_2$ or $MgCO_2$. Many candidates incorrectly gave $H_2$ as a product from an acid–base reaction.

**(e) (i)** Many candidates gave the correct overall dipole for S=C=O but then went on to give the remaining three dipoles incorrectly.

**(ii)** Candidates frequently found it difficult to convert all of the physical quantities to consistent units. Many answers gave the identity as arsenic or some other element and clearly had not used Table 1.2 to identify the compound.

| | | |
|---|---|---|
| 2(a)(i) |  <br>**M1** diagram showing minimum of 4 particles (in total in 2 rows)<br>• circles containing $X^{n+}$ do not have to be labelled<br>•  must be labelled as 'ion' OR empty circles / circles with X must be labelled + ion / positive ion / cation / $X^{n+}$<br>**AND**<br>with the circles surrounded by electrons shown as $e^-$ / – **OR** little circles labelled electrons | **1** |
| | **M2** label / legend showing **delocalised** electrons | **1** |

| | | |
|---|---|---|
| 2(b) | $Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$ | **1** |

| | | |
|---|---|---|
| 2(d)(ii) | moles of $AgNO_3 = \frac{33.65}{1000} \times 0.0500$ $(= 1.68(25) \times 10^{-3})$ | **1** |
| | moles of **X** = ½ × moles of $AgNO_3$ = 0.250 ÷ $M_r$(**X**) $\quad\quad$ ∴ $M_r$(**X**) = 297.2 | **1** |
| | $A_r$ of Group 2 element is 297.2 – 2(79.9) = 137.4 AND **X** is $BaBr_2$ | **1** |

**Question 2**

This was an inorganic-chemistry question based around Group 2.

**(a) (i)** A labelled diagram showing the structure of a metal crystal was required here. Candidates are reminded of the subtle difference between delocalised and free electrons.

**(ii)** There were many good answers here although some responses continued to use free electrons instead of delocalised.

**(b)** A large proportion of candidates offered the reaction with steam ($\rightarrow MgO + H_2$) rather than that with cold water, although there were many good answers.

**(c)** Several answers gave examples of metals which would 'displace' Mg but not Ba. Water was a favourite reagent, with the explanation given as either difference in solubility or a hydrolysis type reaction. Such answers were not sufficiently creditworthy.

**(d) (i)** This question was answered quite well by candidates.

**(ii)** A good proportion of candidates understood what was needed, but many spoilt a good answer by stating that X = Ba. The most common error was not to include the stoichiometric factor of ½: such answers tended to have further inconsistency in the subtraction of the $A_r$ of $1\times$ or $2 \times Cl$.

| | | |
|---|---|---|
| 3(a)(i) | (a reaction where) two (or more) compounds / reagents / molecules form (only) **one product** | **1** |
| 3(a)(ii) | position(al isomerism) | **1** |
| 3(a)(iii) | butanal | **1** |
| 3(a)(iv) | 96% × (5000 / 42.0) × 72.0 = 8230 or 8229 (kg) | **1** |
| 3(b)(iii) | $C_4H_8O + [O] \rightarrow C_4H_8O_2$ | **1** |

**Question 3**

This question was largely framed around organic chemistry, with the inclusion of some physical and stoichiometric work. It was well attempted on the whole, although knowledge of some key definitions was seen to be weaker.

**(a) (i)** Many candidates appreciated that a single product was the outcome of an addition reaction, but many others described/attempted to describe an addition reaction without indicating that only one product was formed.

**(ii)** This was well answered by most candidates.

**(iii)** The major errors here were butan-1-one, butan-1-ol/butanol or propanal/propan-1-al. Candidates are reminded that there is no need to offer a locational number for the aldehyde group.

**(iv)** Very few candidates converted the quantities to moles, so simply offered 96% of 5000 kg as the answer.

**(b) (i)** Candidates seemed largely unaware of the nucleophilic addition mechanism. Common errors and omissions included:

- incorrect placement of the lone pair of electrons on the N of the $CN^-$ ion.
- giving the intermediate a positive charge which was subsequently attacked by the $CN^-$ nucleophile.
- mechanisms involving the H (of HCN) / $H^+$ attacking the intermediate negative ions.

- curly arrows inaccurately coming from the C (in the C=O group).
- the H (of HCN) shown attacking the O (in the C=O group) to form an intermediate hydroxyl derivative with a positive charge.

**(ii)** The first and third marks of this question gave the largest number of variants, with Fehling's reagent or DNPH offered for experiment 1. For experiment 3, many answers erroneously gave a change or loss of colour rather than a method of noting that no reaction had occurred.

**(iii)** There were some very good answers here, although some increased the number of H atoms in the product and/or $H_2$ appeared as a product.

**(iv)** This was generally very well answered, with occasionally some candidates circling two C atoms or the entirety of HOCHCN.

**(c)** It appeared that many candidates did not understand the term molecular formula and gave structural formulae throughout.

**(d) (i)** Some answers referred to a change in position of the equilibrium/yield, which was not credited.

**(ii)** Many answers excluded the word gas and just stated 'less/fewer (number of) moles'. Although the reaction in question was entirely in the gas phase, precision in an answer is crucial. A few candidates gave the incorrect direction of the equilibrium shift.

**(iii)** Many candidates used a value of 436 in their answers, which suggests the use of a Data Booklet. Centres are reminded that Data Booklets are **not** to be issued to candidates for these papers.

**(iv)** Any mention of hydrogen bonding was not credited, in accordance with the convention and definition used by this qualification.